Computational Fluid Dynamics Project Report

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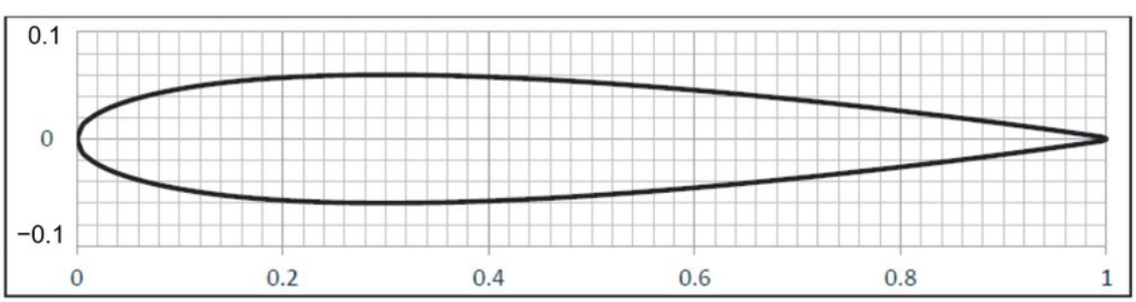
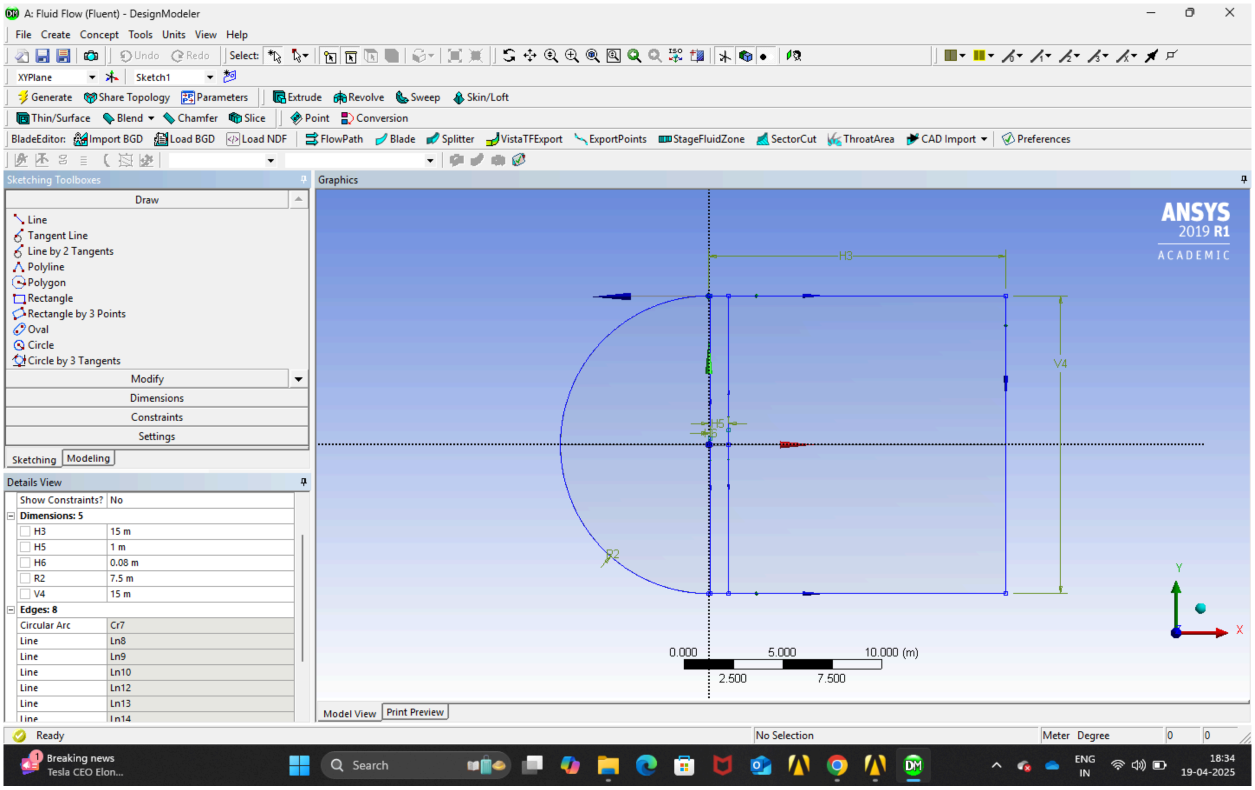
Roll no:- 23110051
Course:- Fluid Dynamics ME-207

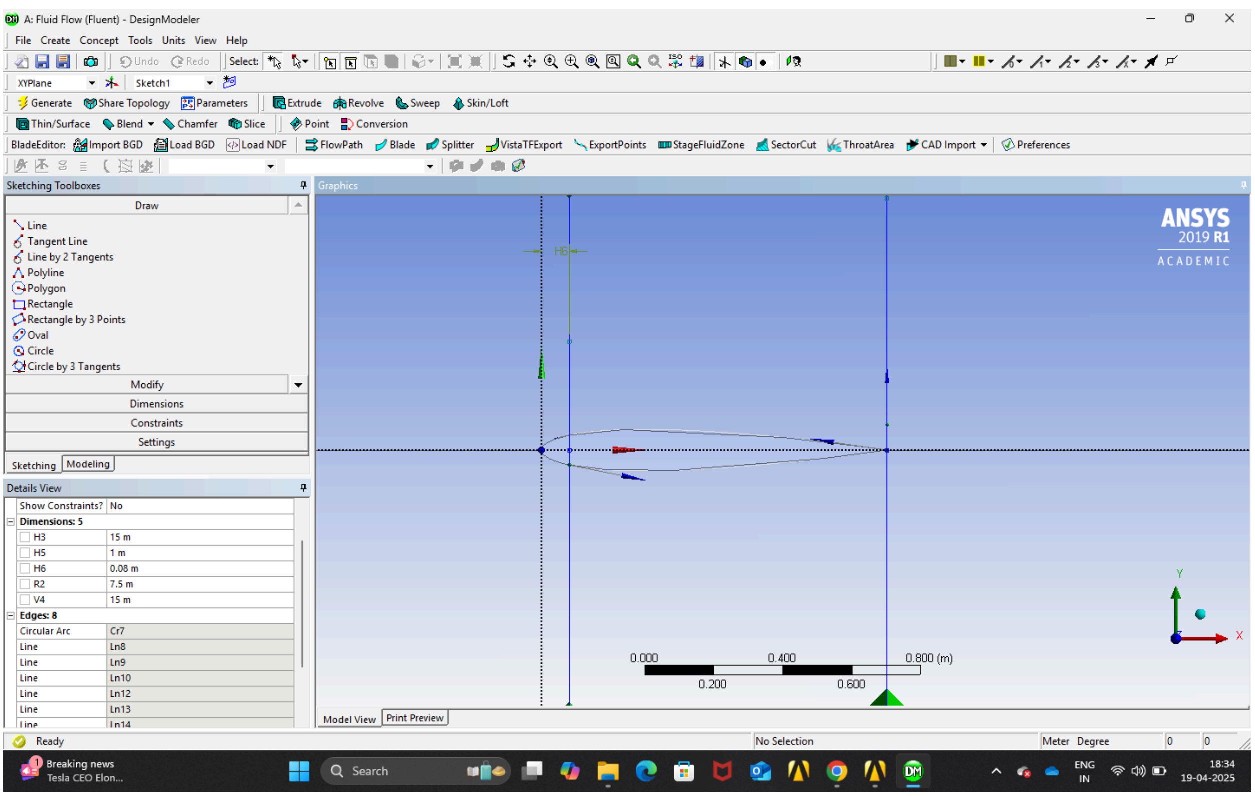
Given:- Freestream velocity - 25 m/s
Temperature - 20 degree celsius, Chord Length - 1m
Note:- Simulation and results have been done Ansys 2019 R1 Academic version

NACA 0012 airfoil section



Geometry


A: Fluid Flow (Fluent) - DesignModeler
Sketching Toolboxes
Draw
Line
Tangent Line
Line by 2 Tangents
Polyline
Polygon
Rectangle
Rectangle by 3 Points
Oval
Circle
Circle by 3 Tangents
Modify
Dimensions
Constraints
Settings
Details View
Show Constraints? No
Dimensions: 5
H3 15 m
H5 1 m
H6 0.08 m
R2 7.5 m
V4 15 m
Edges: 8
Circular Arc Cr7
Line Ln8
Line Ln9
Line Ln10
Line Ln12
Line Ln13
Graphics
ANSYS 2019 R1 ACADEMIC
H3
V4
H5
R2
0.000
5.000
10.000 (m)
2.500
7.500
Ready
No Selection
Meter Degree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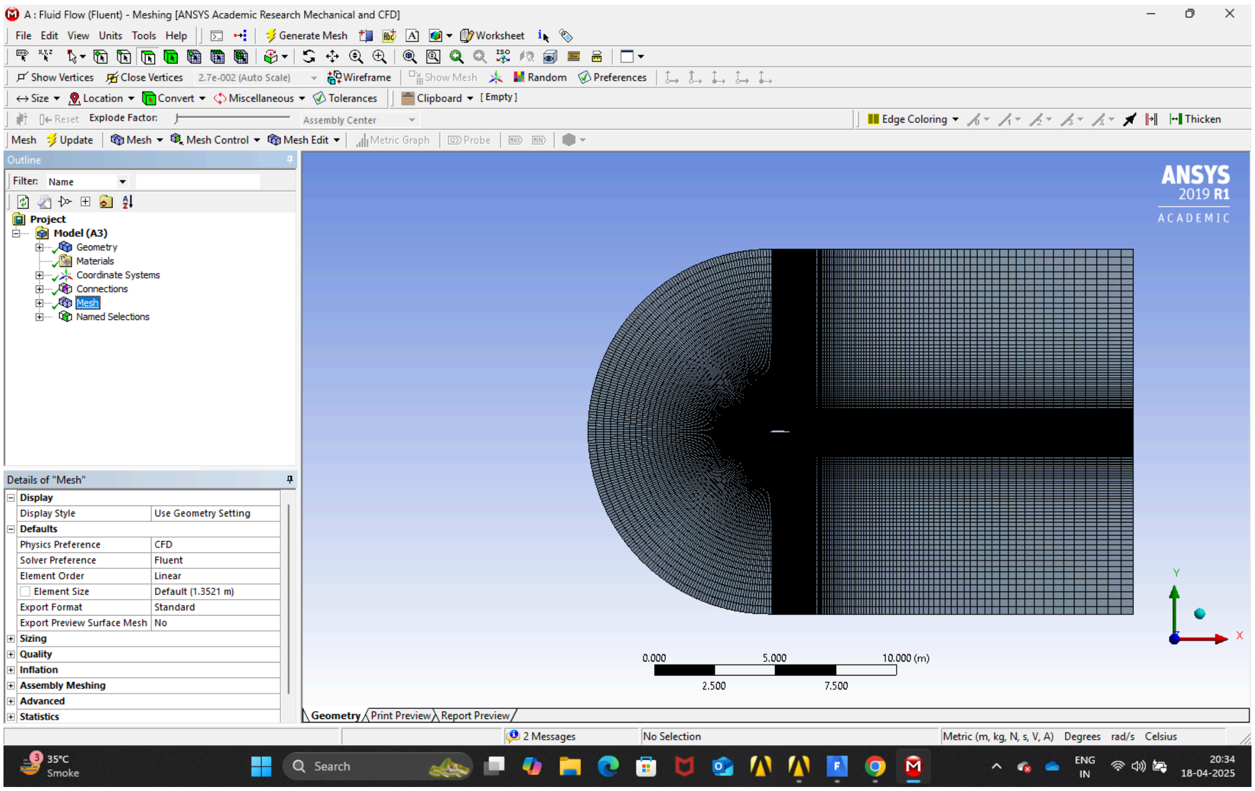
Rectangle
Rectangle by 3 Points
Oval
Circle
Circle by 3 Tangents
Modify
Dimensions
Constraints
Settings
Details View
Show Constraints? No
Dimensions: 5
H3 15 m
H5 1 m
H6 0.08 m
R2 7.5 m
V4 15 m
Edges: 8
Circular Arc Cr7
Line Ln8
Line Ln9
Line Ln10
Line Ln12
Line Ln13
Graphics
ANSYS 2019 R1 ACADEMIC
H6
0.000
0.400
0.800 (m)
0.200
0.600
Ready
No Selection
Meter Degree

Meshing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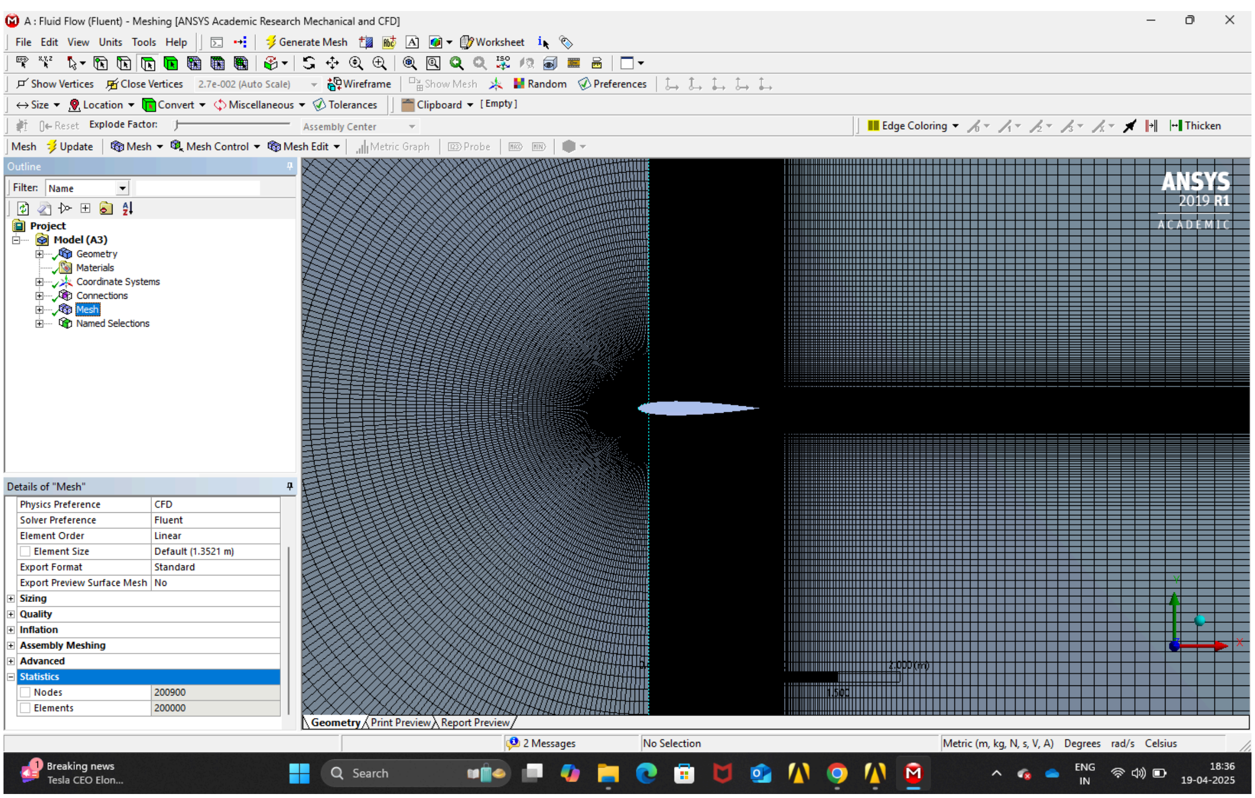
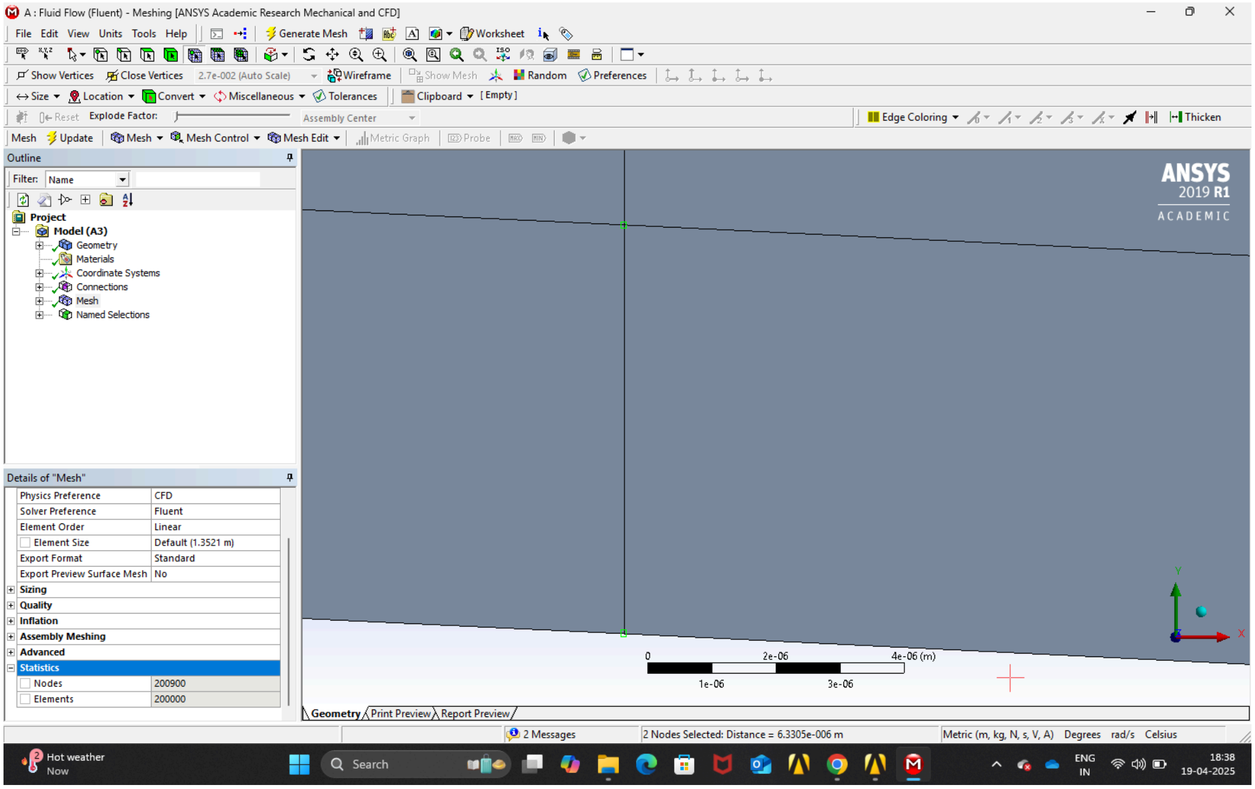
Generate Mesh
Worksheet
Show Vertices
Close Vertices
2.7e-002 (Auto Scale)
Wireframe
Show Mesh
Random
Preferences
Size
Location
Convert
Miscellaneous
Tolerances
Clipboard
[Empty]
Reset
Explode Factor:
Assembly Center
Edge Coloring
Thicken
Mesh
Update
Mesh Control
Mesh Edit
Metric Graph
Probe
Outline
Filter: Name
Project
Model (A3)
Geometry
Materials
Coordinate Systems
Connections
Mesh
Named Selections
Details of "Mesh"
Display
Display Style
Use Geometry Setting
Defaults
Physics Preference
CFD
Solver Preference
Fluent
Element Order
Linear
Element Size
Default (1.3521 m)
Export Format
Standard
Export Preview Surface Mesh
No
Sizing
Quality
Inflation
Assembly Meshing
Advanced
Statistics
ANSYS
2019 R1
ACADEMIC
0.000
5.000
10.000 (m)
2.500
7.500
Y
X
Geometry
Print Preview
Report Preview
2 Messages
No Selection
Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
35°C
Smoke
Search
ENG
IN
20:34
18-04-2025

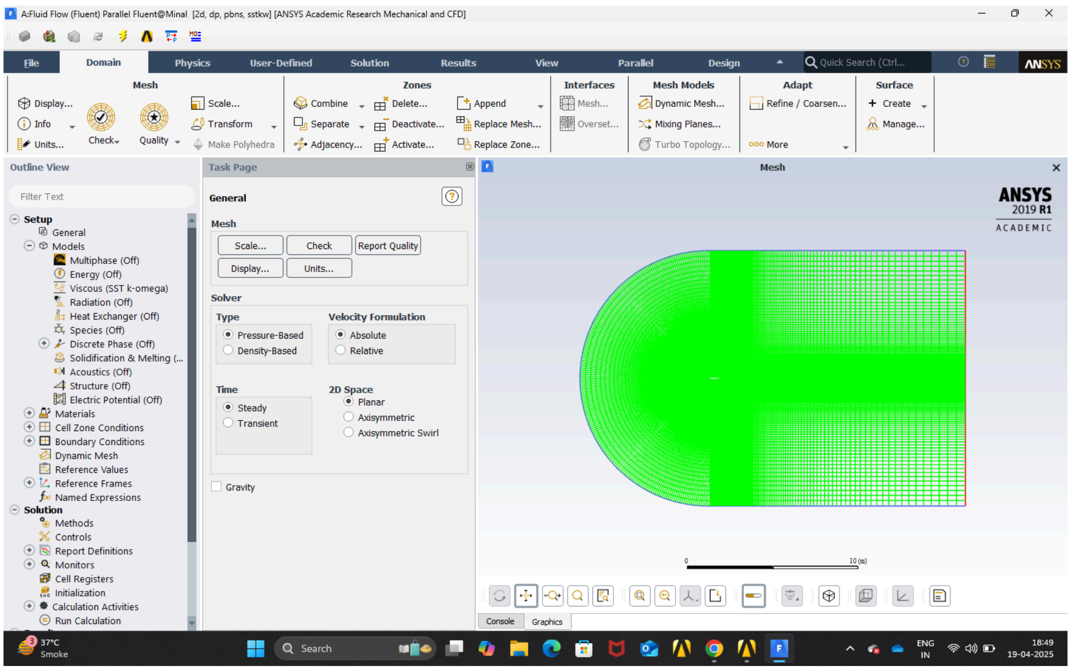

A : Fluid Flow (Fluent) - Meshing [ANSYS Academic Research Mechanical and CFD]
File Edit View Units Tools Help
Generate Mesh
Worksheet
Show Vertices
Close Vertices
2.7e-002 (Auto Scale)
Wireframe
Show Mesh
Random
Preferences
Size
Location
Convert
Miscellaneous
Tolerances
Clipboard
[Empty]
Reset
Explode Factor:
Assembly Center
Edge Coloring
Thicken
Mesh
Update
Mesh Control
Mesh Edit
Metric Graph
Probe
Outline
Filter: Name
Project
Model (A3)
Geometry
Materials
Coordinate Systems
Connections
Mesh
Named Selections
Details of "Mesh"
Physics Preference
CFD
Solver Preference
Fluent
Element Order
Linear
Element Size
Default (1.3521 m)
Export Format
Standard
Export Preview Surface Mesh
No
Sizing
Quality
Inflation
Assembly Meshing
Advanced
Statistics
Nodes
200900
Elements
200000
ANSYS
2019 R1
ACADEMIC
Geometry
Print Preview
Report Preview
2 Messages
No Selection
Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Breaking news
Tesla CEO Elon...
Search
ENG
IN
18:36
19-04-2025

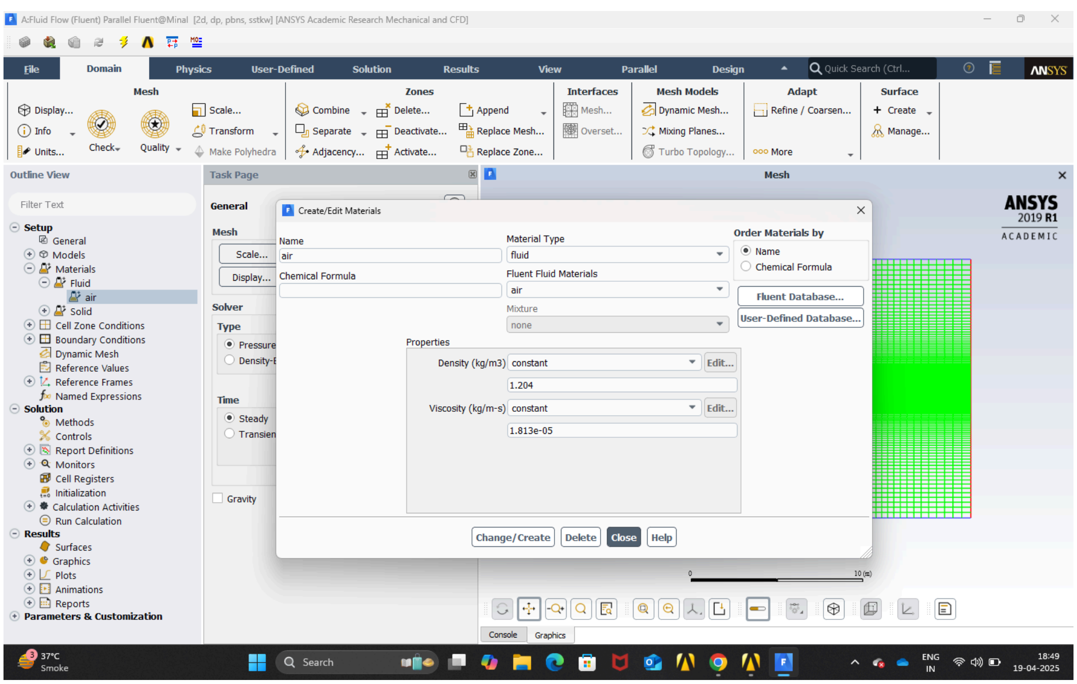


- The total number of nodes = 200900
- The total number of elements = 200000
- The distance between two nodes = 6.3305e-006 m which gives y+ less than 1, hence the results obtained by the mesh obtained will be accurate and all the boundary effects will be captured accurately.
- The meshing near the airfoil is more concentrated because the airfoil is the region of our interest and if the meshing is not good enough then the results will vary significantly.
- Formula for calculating Reynold's number is $Re = (\rho v L) / \mu$
  - ρ is the density
  - V is the velocity
  - L is the length
  - μ is the dynamic viscosity

Discretisation schemes and solution methodology (The setup is for Angle of attack of 5º)

The Reynold's number for the flow is approximately 1.6 x 10^6 so the flow is turbulent as the transitional flow is around Reynolds' number is approximately 10^5, so the laminar model will not give accurate results so we have to use the SST-K Omega model to get correct results.

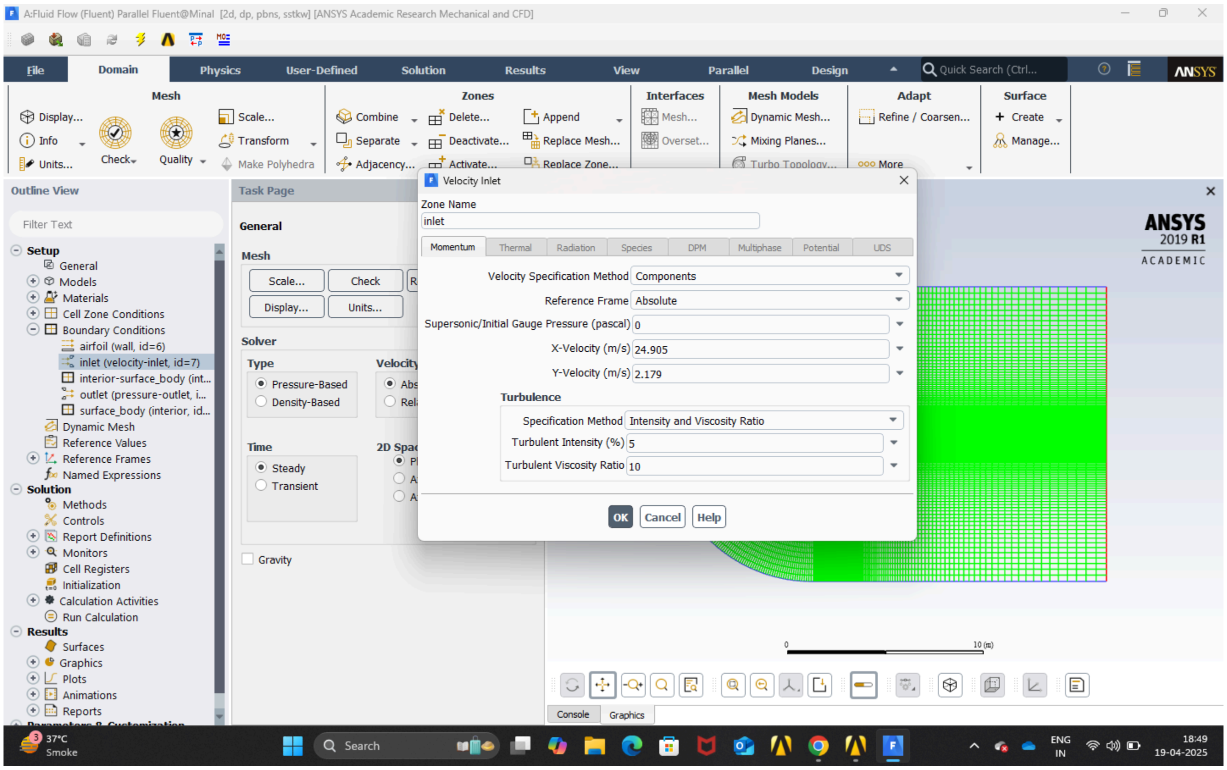


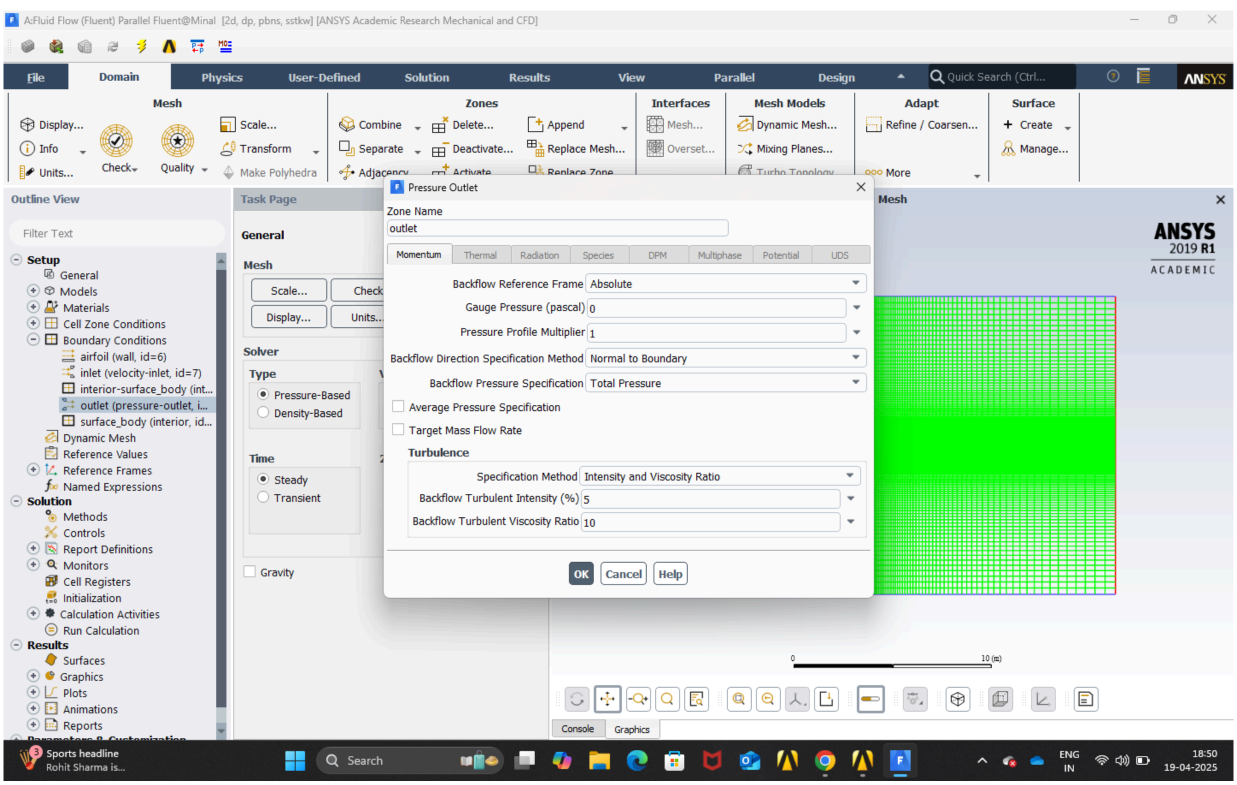


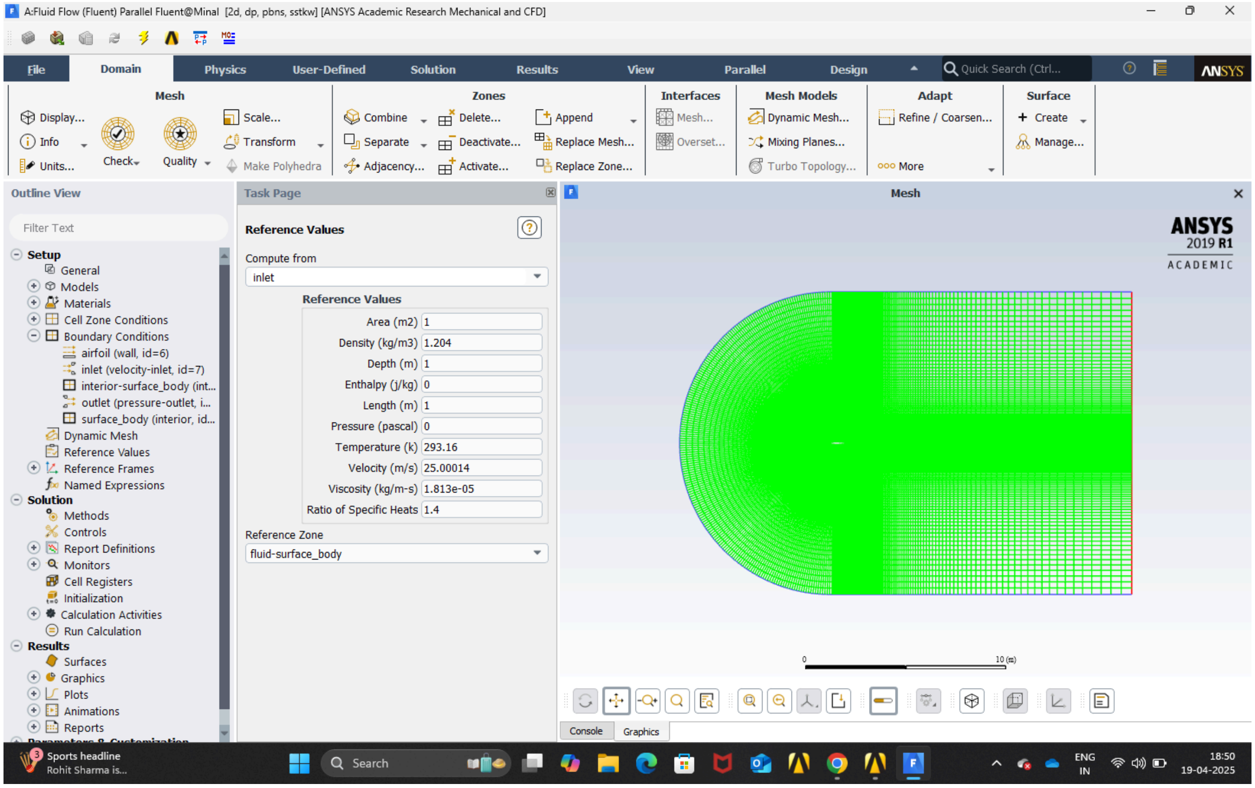
The viscosity and density values are taken at 20ºc.

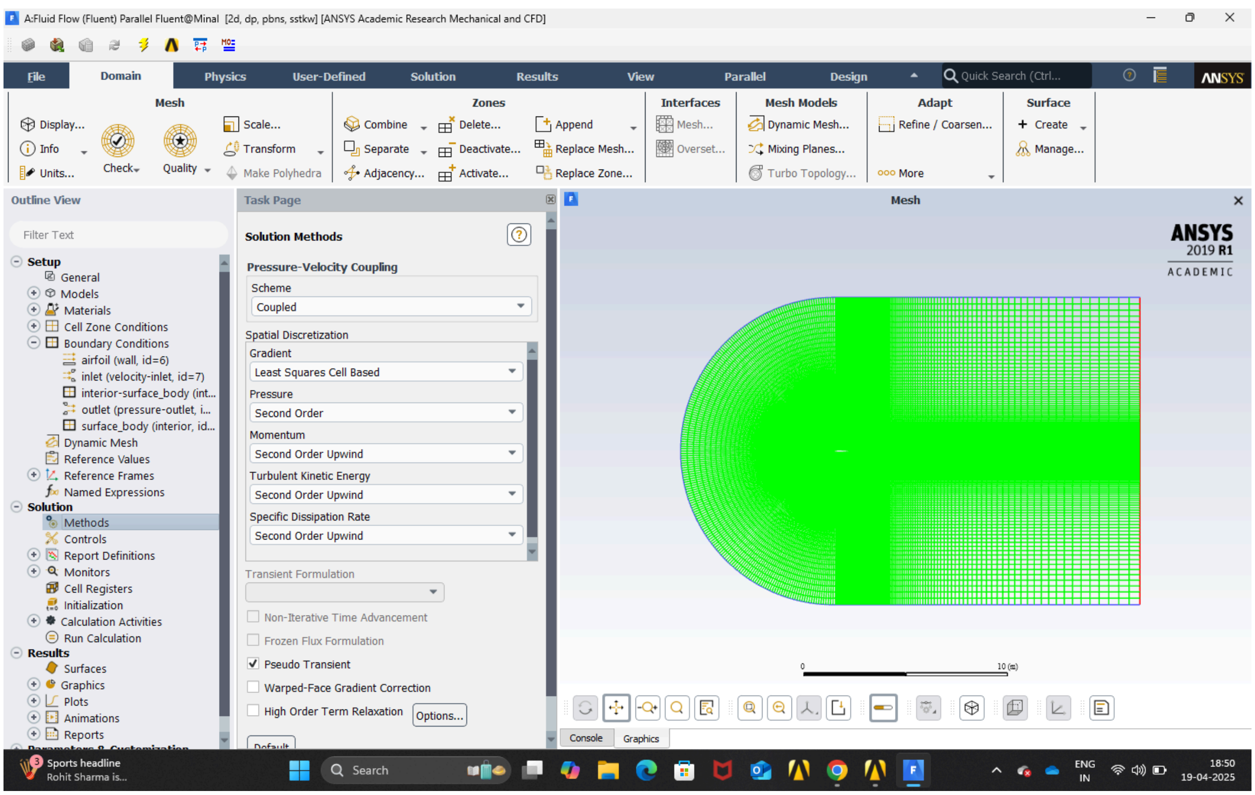
The velocities are broken into x and y components for 5° angle of attack.

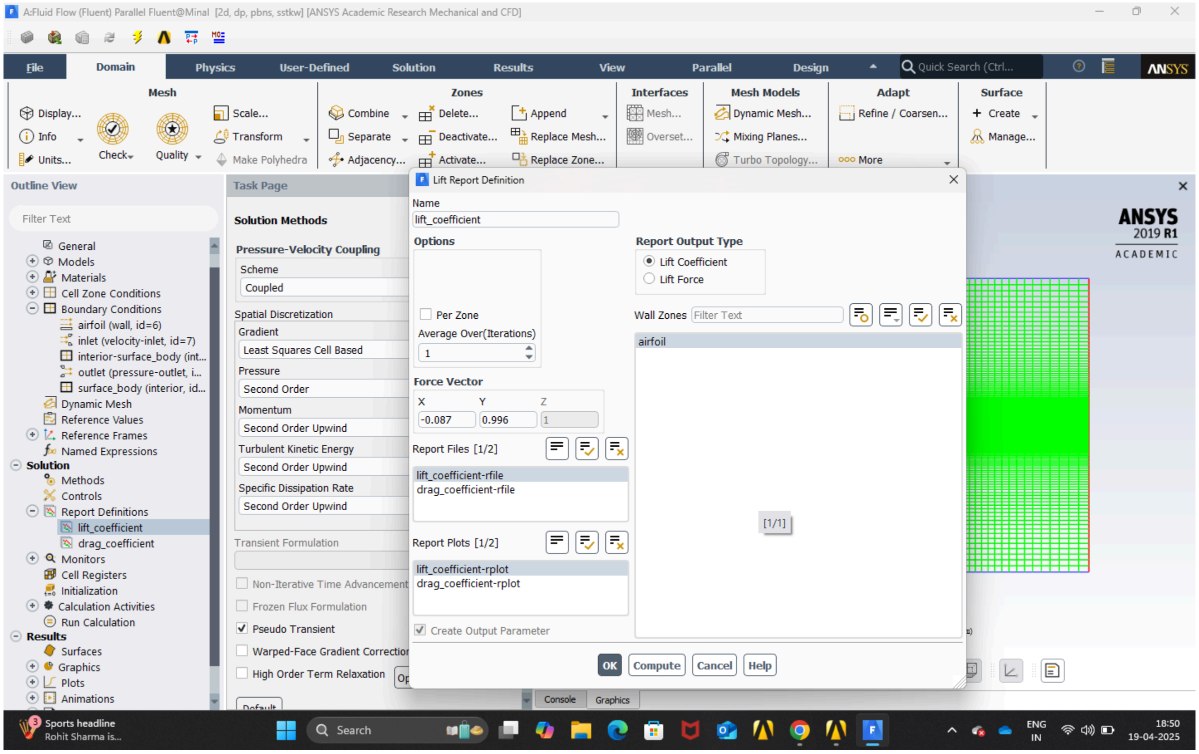
$V_x = V \cos 5^{\circ}$ & $V_y = V \sin 5^{\circ}$

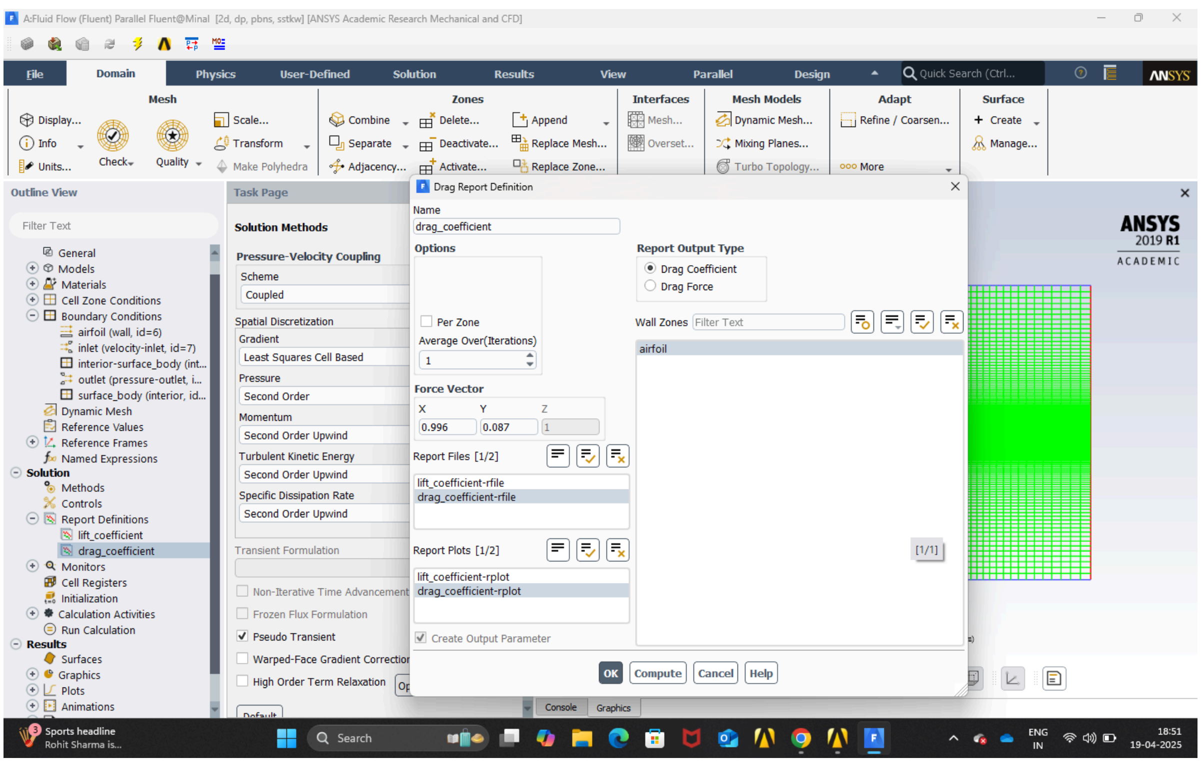




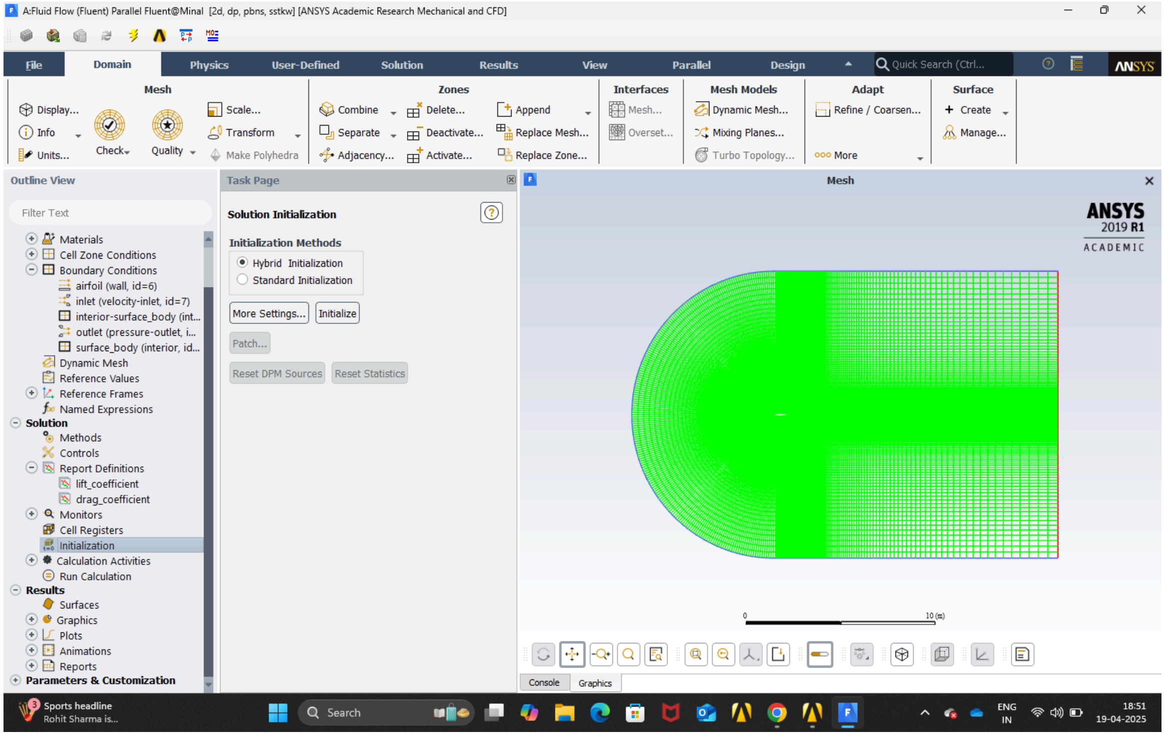


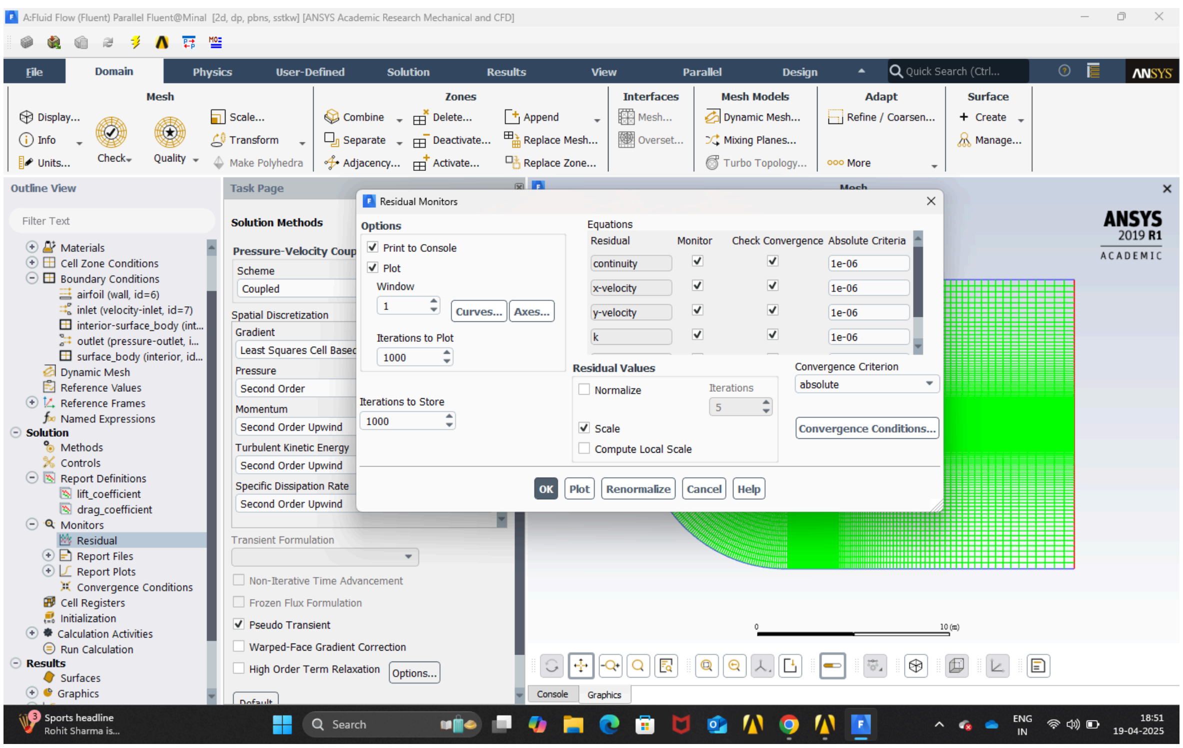


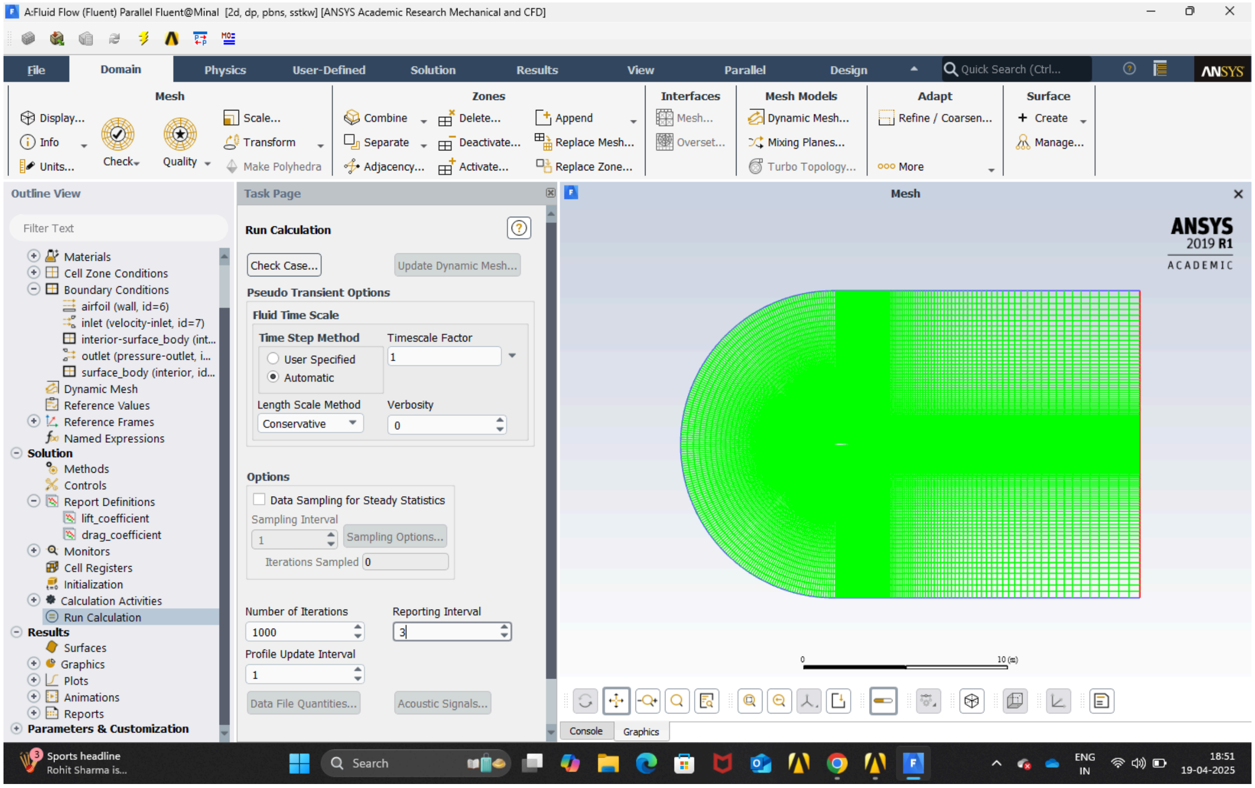




This step is done to plot the lift and drag coefficient graph and the force vector changes with change in Angle of attack and it is different for drag coefficient and lift coefficient.


A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
File
Domain
Physics
User-Defined
Solution
Results
View
Parallel
Design
Outline View
Task Page
Solution Initialization
Initialization Methods
Hybrid Initialization
Standard Initialization
More Settings...
Initialize
Patch...
Reset DPM Sources
Reset Statistics
Mesh
ANSYS
2019 R1
ACADEMIC
Materials
Cell Zone Conditions
Boundary Conditions
airfoil (wall, id=6)
inlet (velocity-inlet, id=7)
interior-surface_body (int...
outlet (pressure-outlet, i...
surface_body (interior, id...
Dynamic Mesh
Reference Values
Reference Frames
Named Expressions
Solution
Methods
Controls
Report Definitions
lift_coefficient
drag_coefficient
Monitors
Cell Registers
Initialization
Calculation Activities
Run Calculation
Results
Surfaces
Graphics
Plots
Animations
Reports
Parameters & Customization
Console
Graphics
18:51
19-04-2025

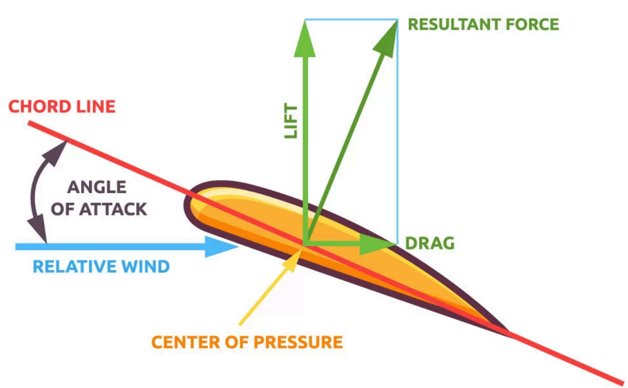

A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Task Page
Solution Methods
Pressure-Velocity Coup
Scheme
Coupled
Spatial Discretization
Gradient
Least Squares Cell Based
Pressure
Second Order
Momentum
Second Order Upwind
Turbulent Kinetic Energy
Second Order Upwind
Specific Dissipation Rate
Second Order Upwind
Transient Formulation
Non-Iterative Time Advancement
Frozen Flux Formulation
Pseudo Transient
Warped-Face Gradient Correction
High Order Term Relaxation
Options...
Residual Monitors
Options
Print to Console
Plot
Window
1
Curves...
Axes...
Iterations to Plot
1000
Iterations to Store
1000
Equations
Residual
Monitor
Check Convergence
Absolute Criteria
continuity
1e-06
x-velocity
1e-06
y-velocity
1e-06
k
1e-06
Residual Values
Normalize
Iterations
5
Scale
Compute Local Scale
Convergence Criterion
absolute
Convergence Conditions...
OK
Plot
Renormalize
Cancel
Help
Monitors
Residual
Report Files
Report Plots
Convergence Conditions
ANSYS
2019 R1
ACADEMIC
18:51
19-04-2025



Final step is to run the calculation and the number of iterations is set 1000.

- For 0º angle of attack the only changes have to be made to the velocity components because now there will be x - component of velocity and the next change will be change to the drag and lift coefficient force vector which will now be (X - 1, Y - 0, Z - 0) for drag coefficient and (X - , Y - 1, Z - 0) for lift coefficient.

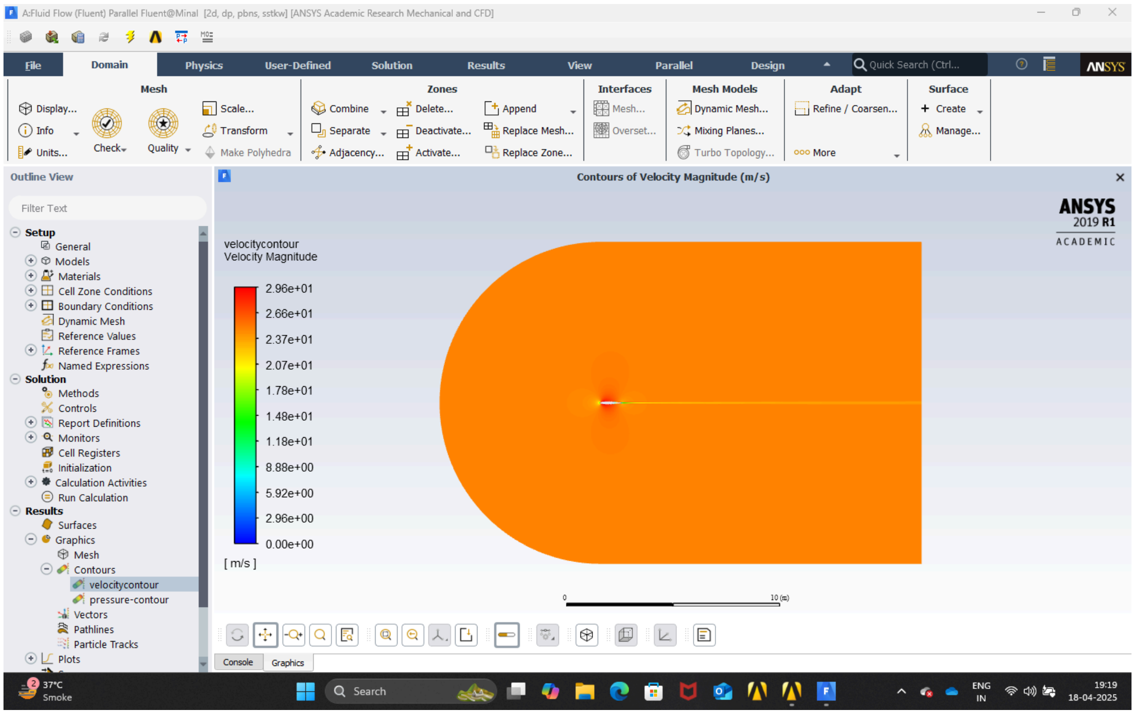
- For 5º angle of attack the change to the drag and lift coefficient force vector was made because they are measured parallel and normal to the flow directions respectively.

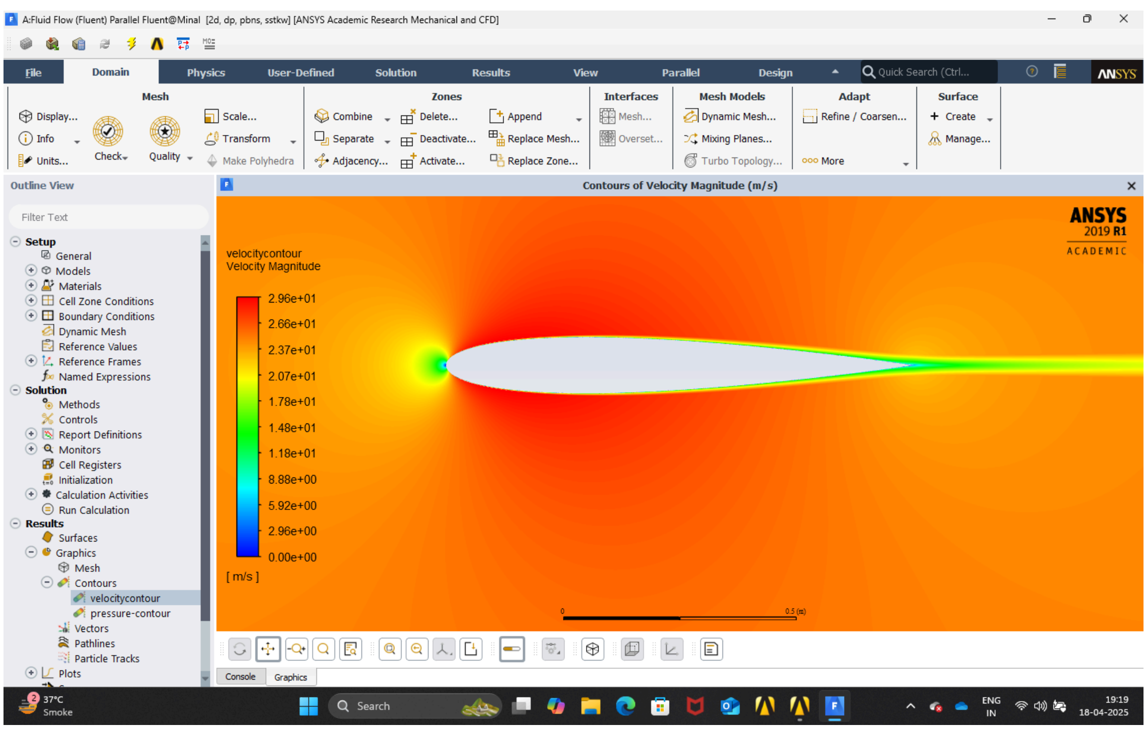


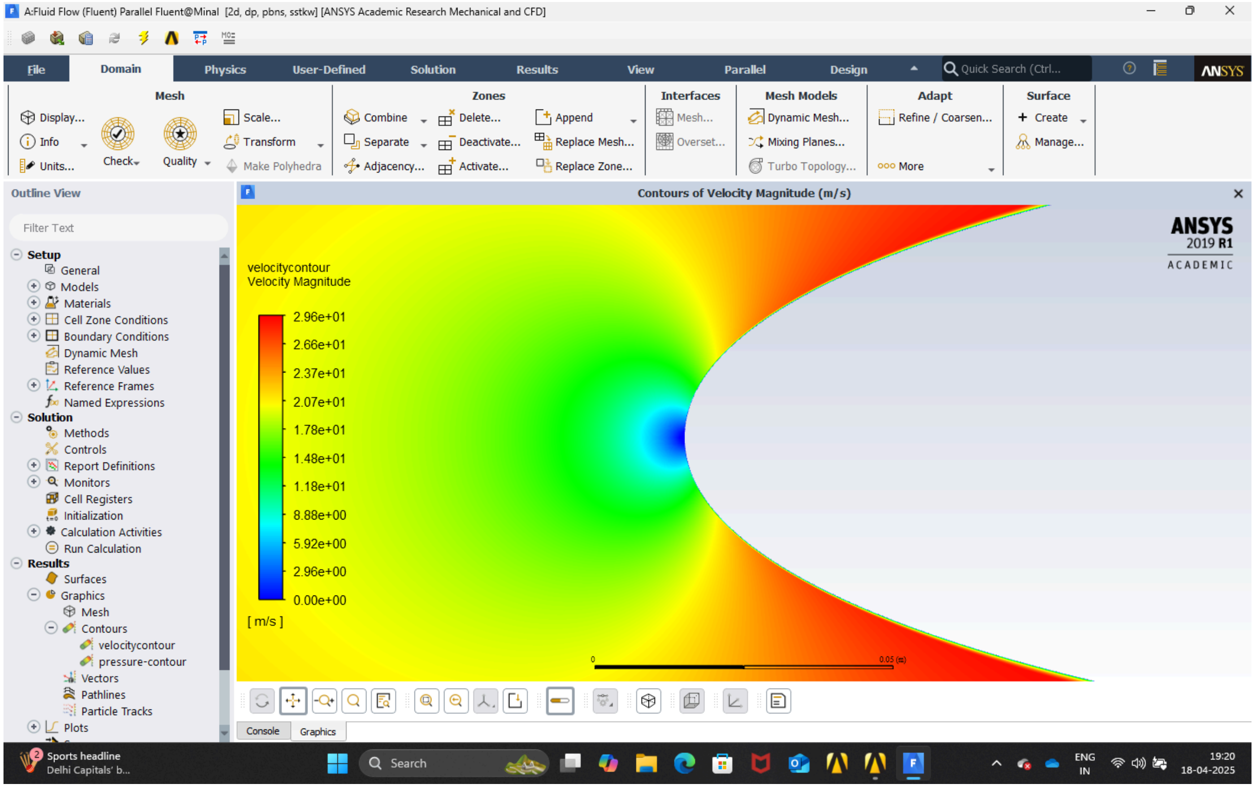
Results and Discussions

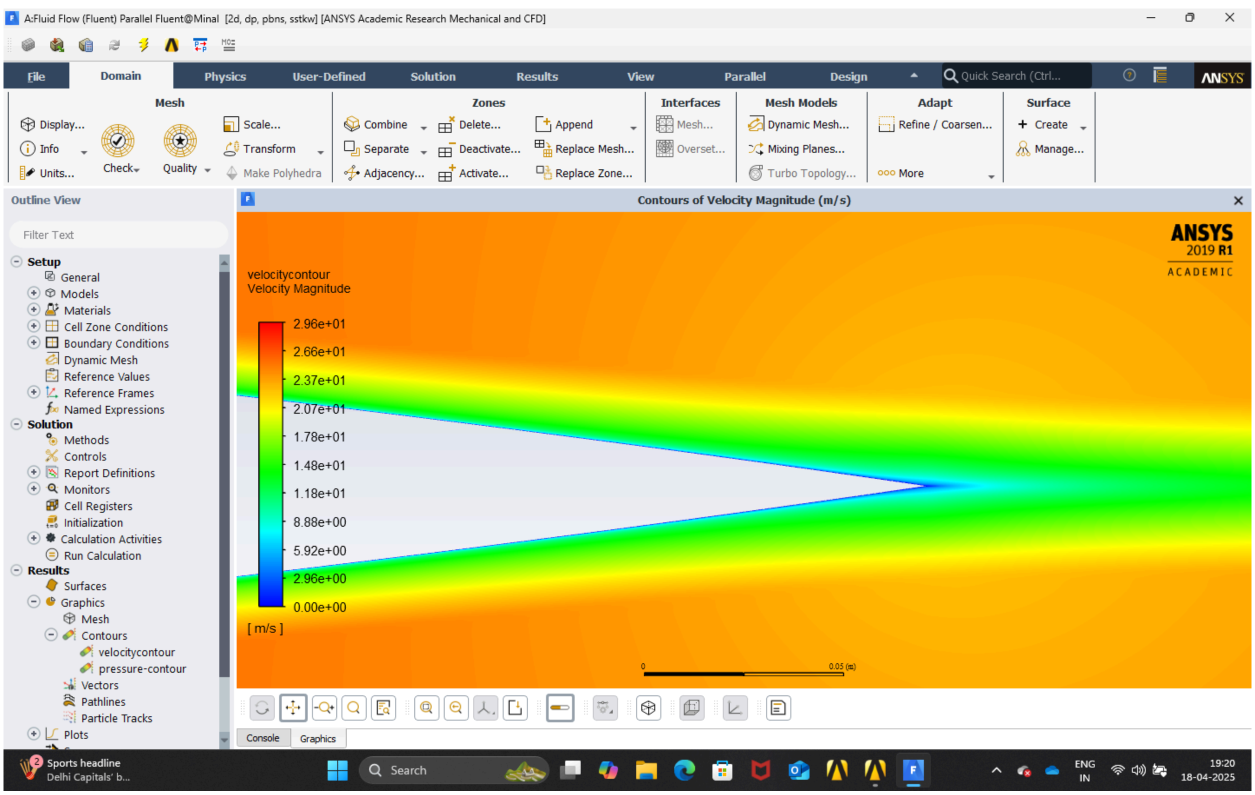
Case - 1 Angle of attack = 0 degrees

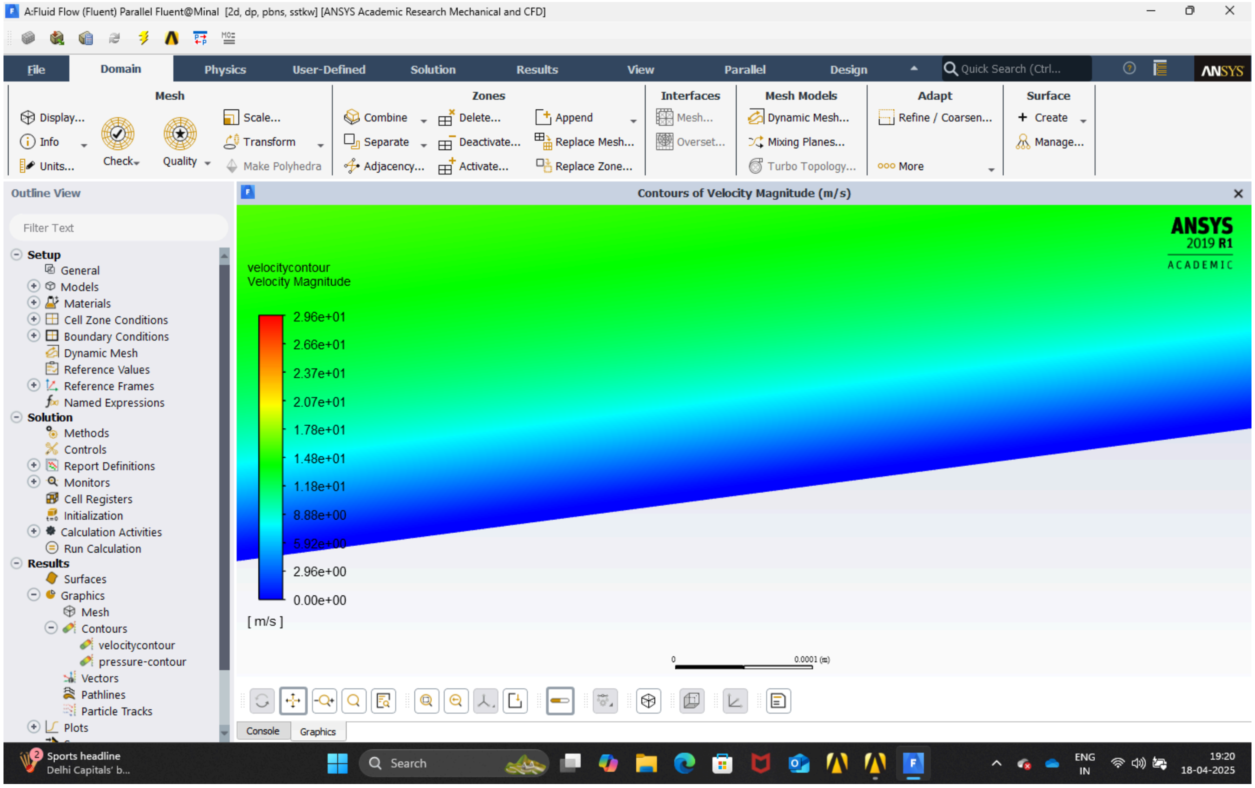
Velocity Contours






A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Contours of Velocity Magnitude (m/s)
velocitycontour
Velocity Magnitude
2.96e+01
2.66e+01
2.37e+01
2.07e+01
1.78e+01
1.48e+01
1.18e+01
8.88e+00
5.92e+00
2.96e+00
0.00e+00
[m/s]
ANSYS
2019 R1
ACADEMIC


A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Contours of Velocity Magnitude (m/s)
velocitycontour
Velocity Magnitude
2.96e+01
2.66e+01
2.37e+01
2.07e+01
1.78e+01
1.48e+01
1.18e+01
8.88e+00
5.92e+00
2.96e+00
0.00e+00
[m/s]
ANSYS
2019 R1
ACADEMIC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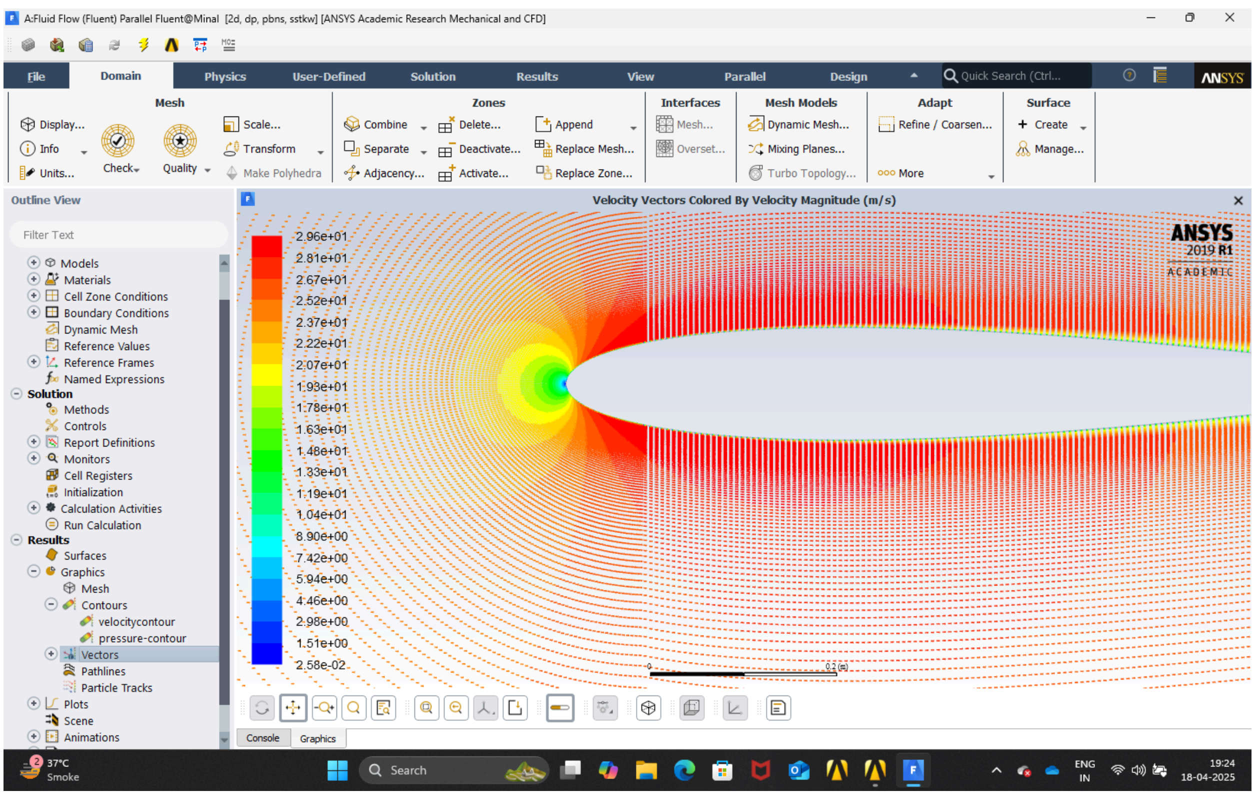
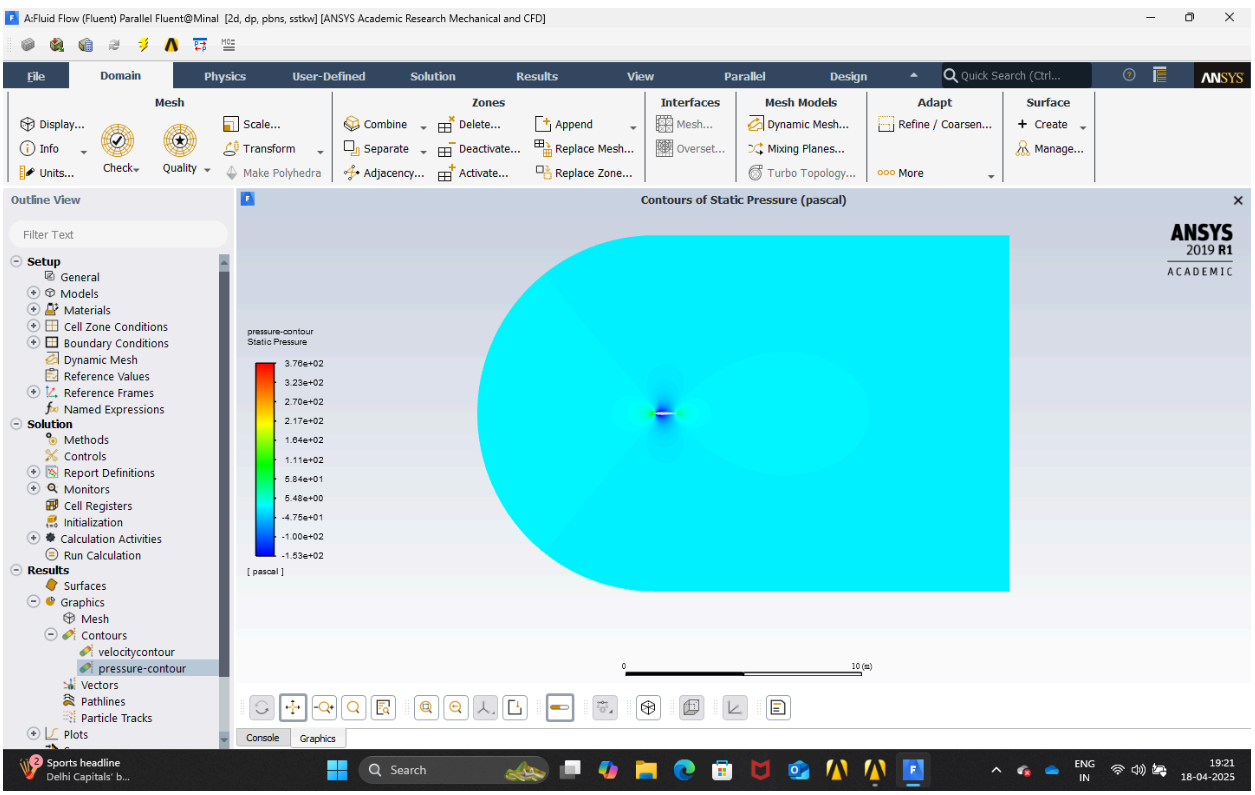
Contours of Velocity Magnitude (m/s)
velocitycontour
Velocity Magnitude
2.96e+01
2.66e+01
2.37e+01
2.07e+01
1.78e+01
1.48e+01
1.18e+01
8.88e+00
5.92e+00
2.96e+00
0.00e+00
[m/s]
ANSYS
2019 R1
ACADEMIC
Outline View
Setup
General
Models
Materials
Cell Zone Conditions
Boundary Conditions
Dynamic Mesh
Reference Values
Reference Frames
Named Expressions
Solution
Methods
Controls
Report Definitions
Monitors
Cell Registers
Initialization
Calculation Activities
Run Calculation
Results
Surfaces
Graphics
Mesh
Contours
velocitycontour
pressure-contour
Vectors
Pathlines
Particle Tracks
Plots
Console
Graphics
19:20
18-04-2025

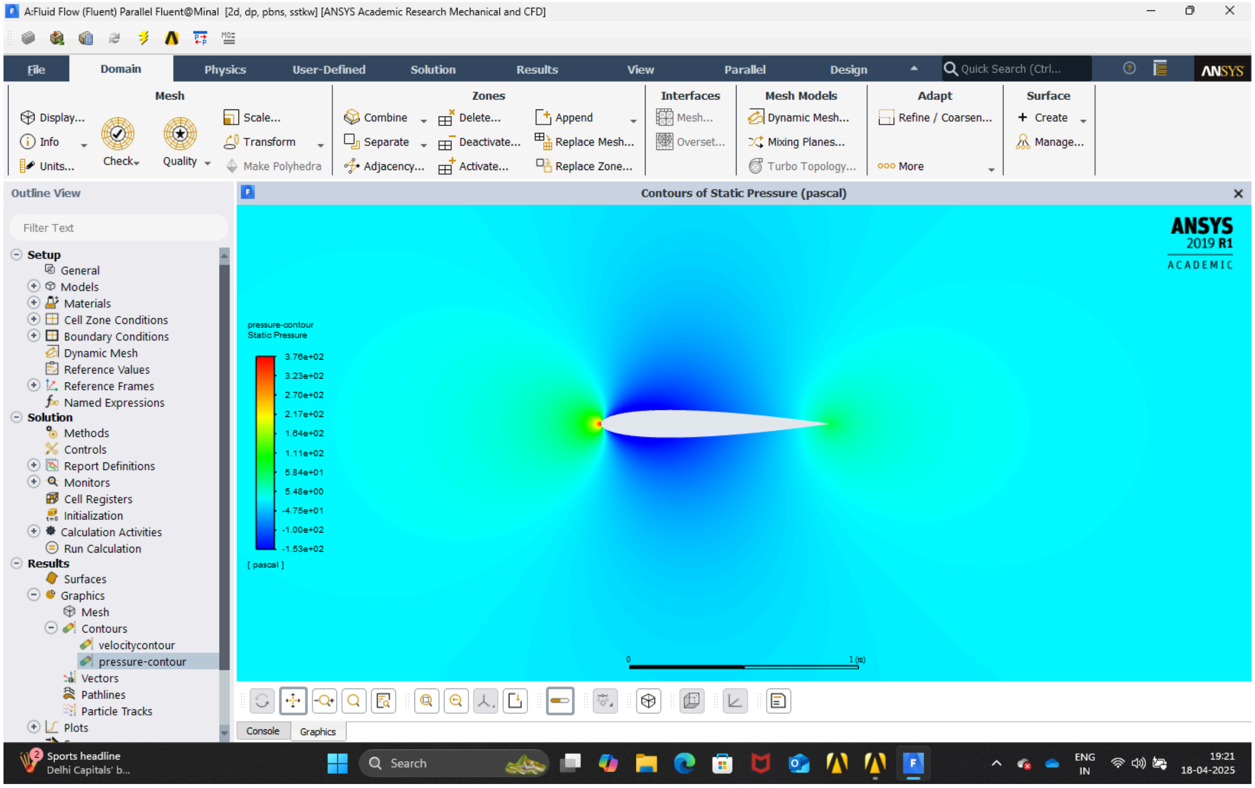

A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Velocity Vectors Colored By Velocity Magnitude (m/s)
2.96e+01
2.81e+01
2.67e+01
2.52e+01
2.37e+01
2.22e+01
2.07e+01
1.93e+01
1.78e+01
1.63e+01
1.48e+01
1.33e+01
1.19e+01
1.04e+01
8.90e+00
7.42e+00
5.94e+00
4.46e+00
2.98e+00
1.51e+00
2.58e-02
ANSYS
2019 R1
ACADEMIC
Outline View
Models
Materials
Cell Zone Conditions
Boundary Conditions
Dynamic Mesh
Reference Values
Reference Frames
Named Expressions
Solution
Methods
Controls
Report Definitions
Monitors
Cell Registers
Initialization
Calculation Activities
Run Calculation
Results
Surfaces
Graphics
Mesh
Contours
velocitycontour
pressure-contour
Vectors
Pathlines
Particle Tracks
Plots
Scene
Animations
Console
Graphics
19:24
18-04-2025

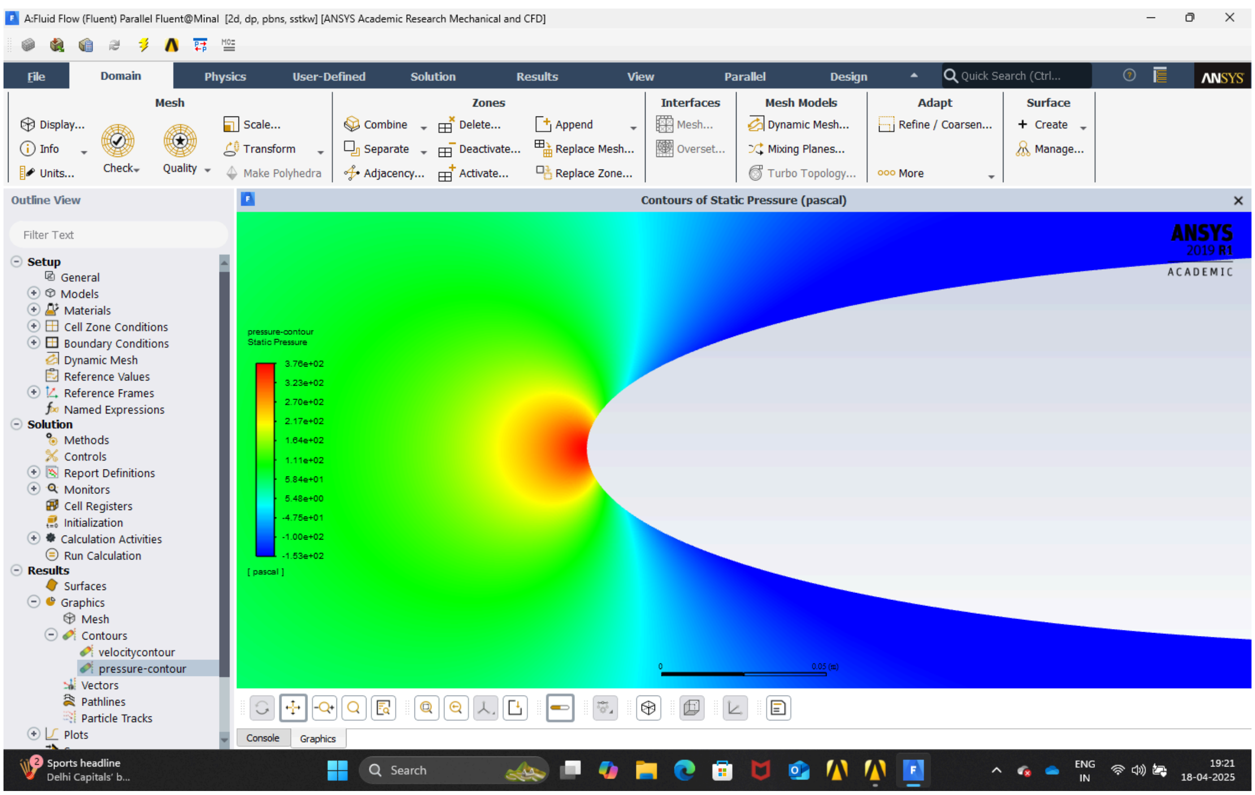
Observations

- The dark blue colour on the walls represents the no slip condition.
- The contours are symmetric as the angle of attack is zero degrees and the airfoil is symmetric.
- At the leading edge of the airfoil the velocity is zero indicated by dark blue showing stagnation point.
- Just above and below the walls the velocity reaches its maximum value.

Pressure Contour




A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Contours of Static Pressure (pascal)
pressure-contour
Static Pressure
3.76e+02
3.23e+02
2.70e+02
2.17e+02
1.64e+02
1.11e+02
5.84e+01
5.48e+00
-4.75e+01
-1.00e+02
-1.53e+02
[pascal]
ANSYS
2019 R1
ACADEMIC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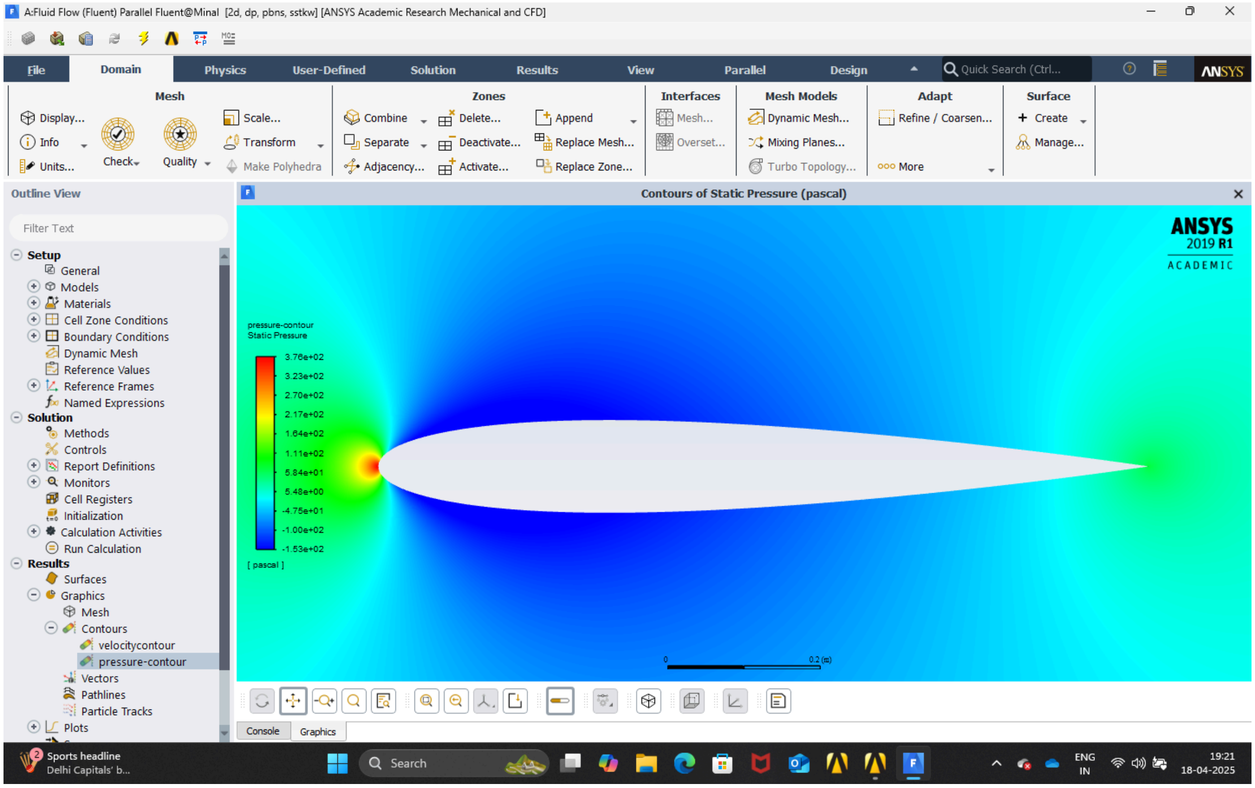
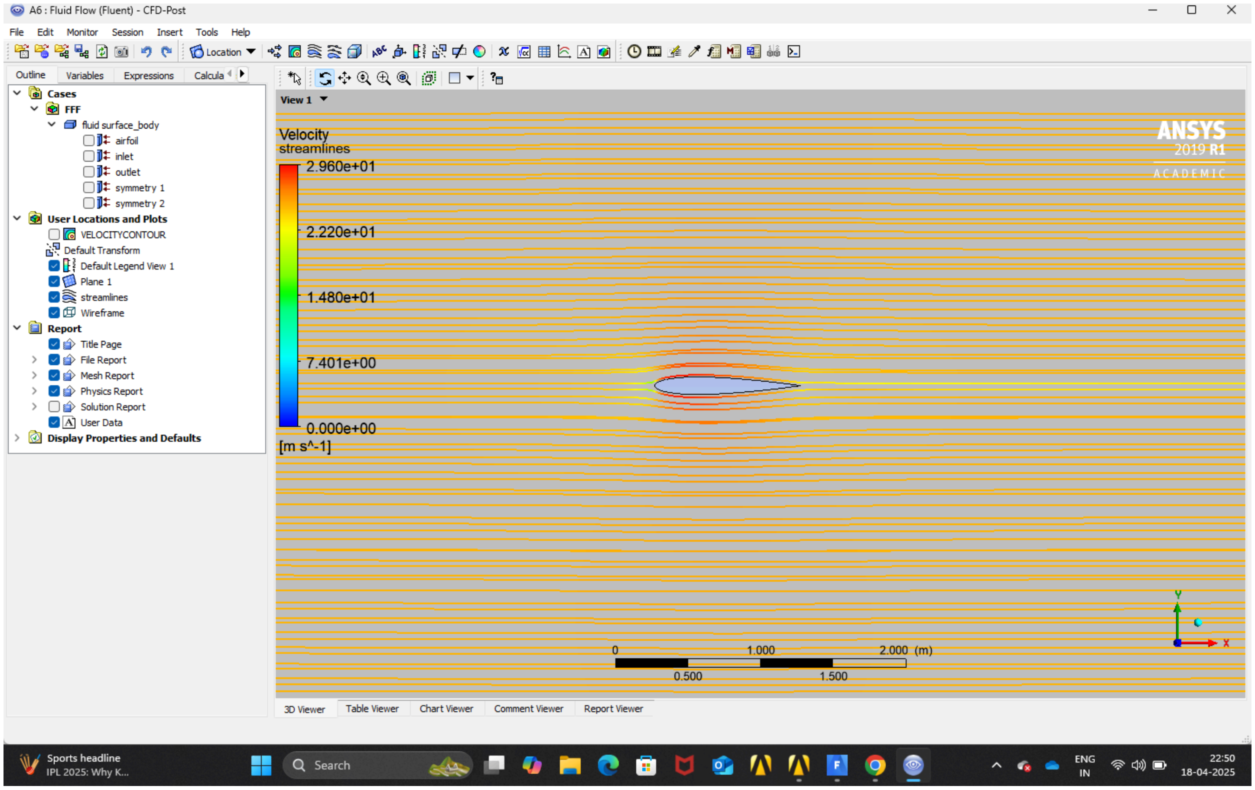
Contours of Static Pressure (pascal)
pressure-contour
Static Pressure
3.76e+02
3.23e+02
2.70e+02
2.17e+02
1.64e+02
1.11e+02
5.84e+01
5.48e+00
-4.75e+01
-1.00e+02
-1.53e+02
[pascal]
ANSYS
2019 R1
ACADEMIC

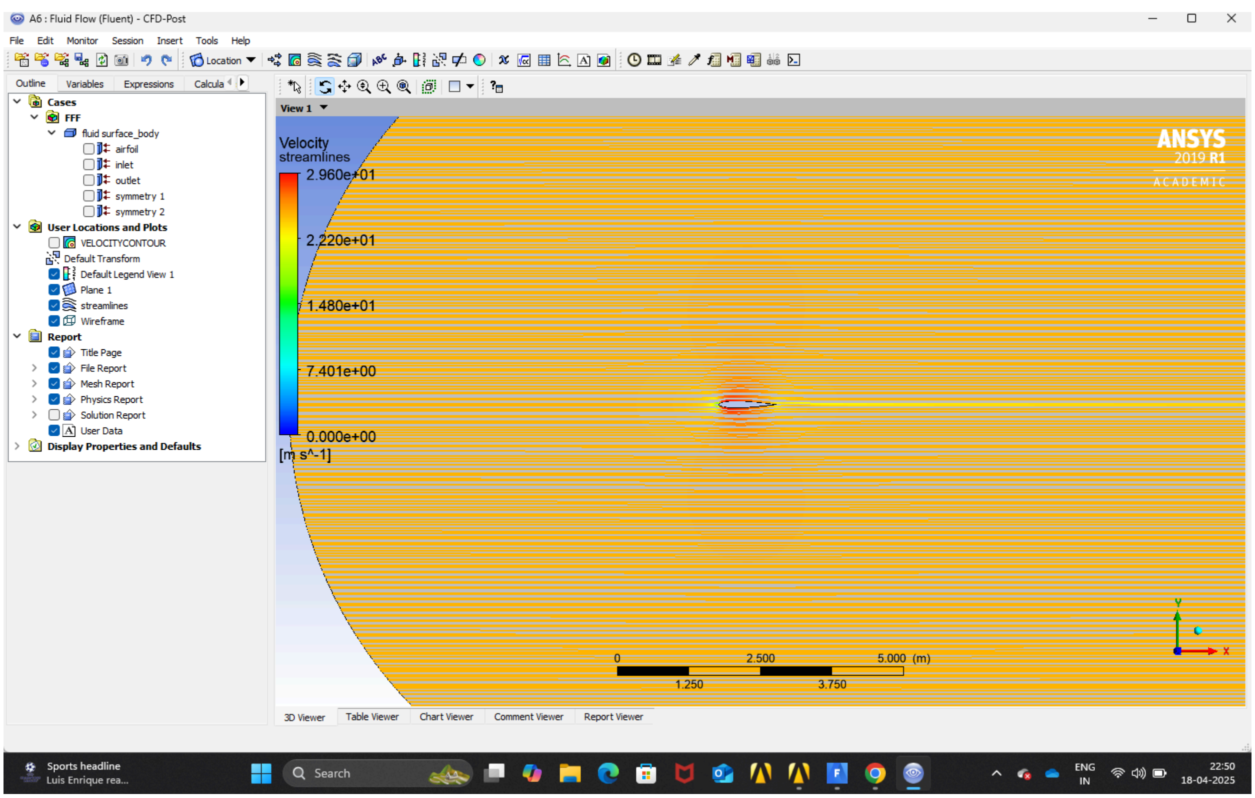


<u>Observations</u>

- The pressure is maximum at the wall following the no slip condition.
- The contours are symmetric as the angle of attack is zero degrees and the airfoil is symmetric.
- At the leading edge of the airfoil the pressure is maximum indicated by dark red showing stagnation point following Bernoulli's principle.
- Just above and below the walls the pressure reaches its minimum value also following Bernoulli’s principle.

Streamlines



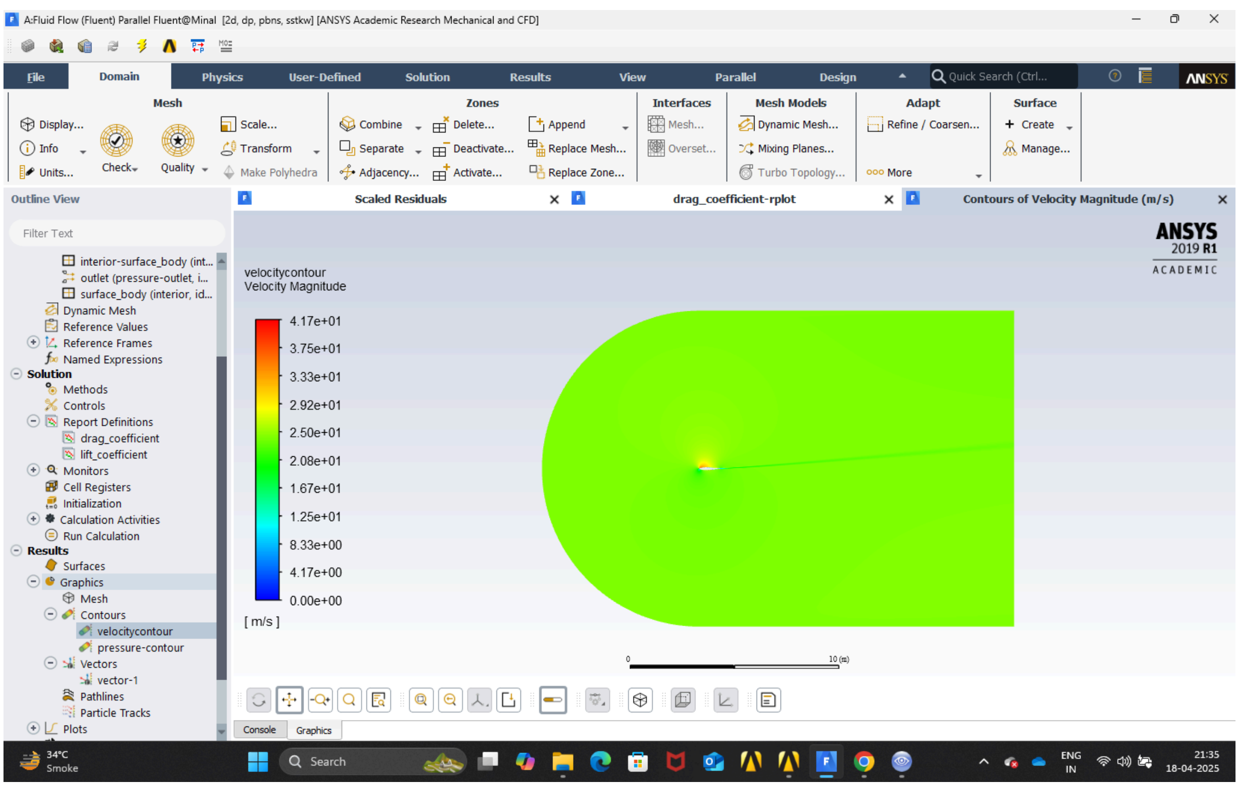


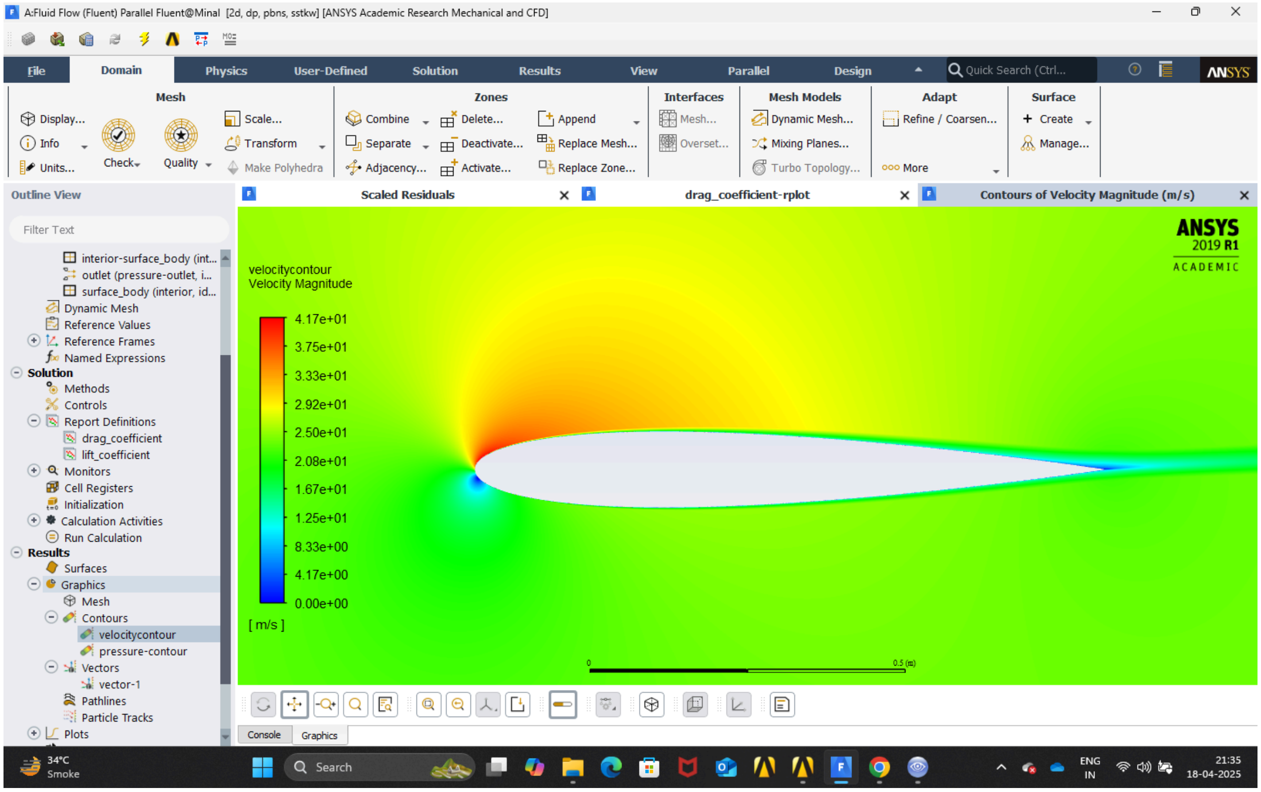
Observations

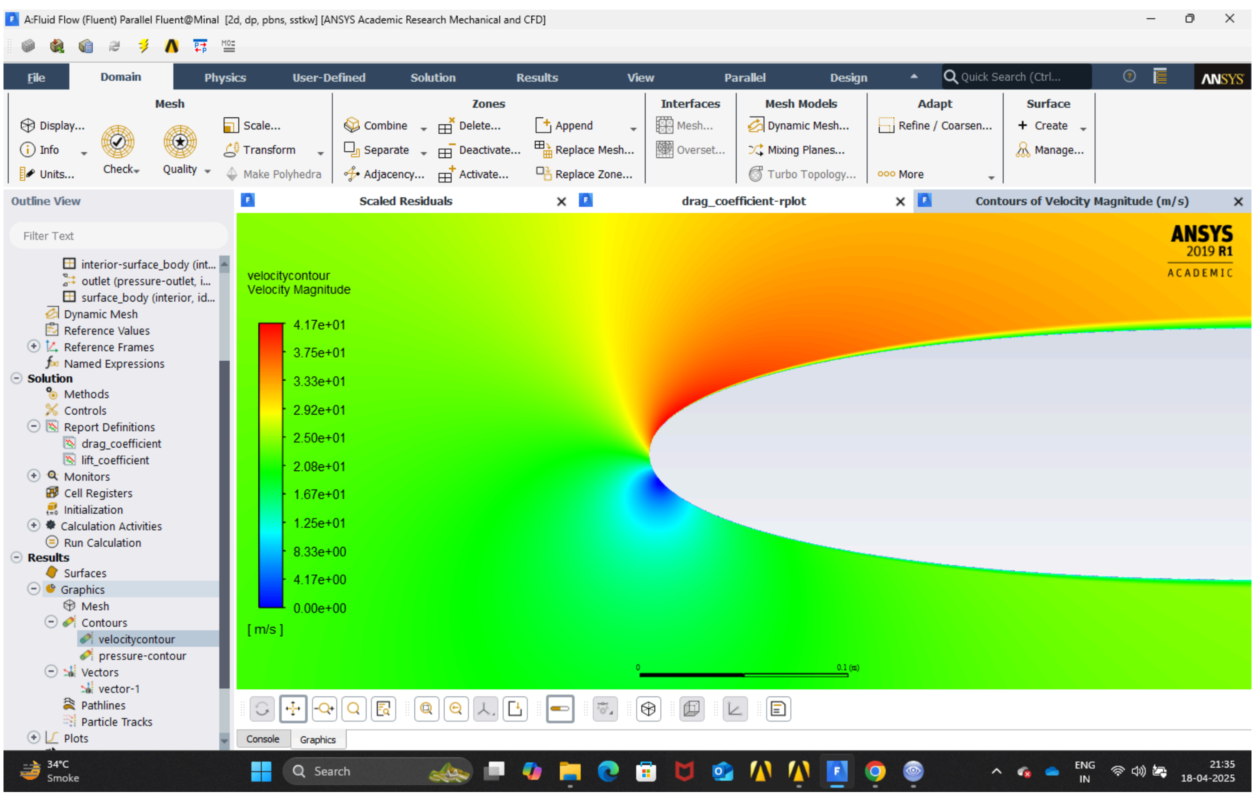
- The streamlines faraway from the airfoil are mostly a horizontal line as the flow is only in the x - direction.
- Near the airfoil we can observe the streamline flowing over it as the streamlines are curved.
- The velocity increases from the leading edge of the airfoil to the middle and then again decelerates and it is indicated by the streamlines getting to a color of dark red and then again going back to yellow.

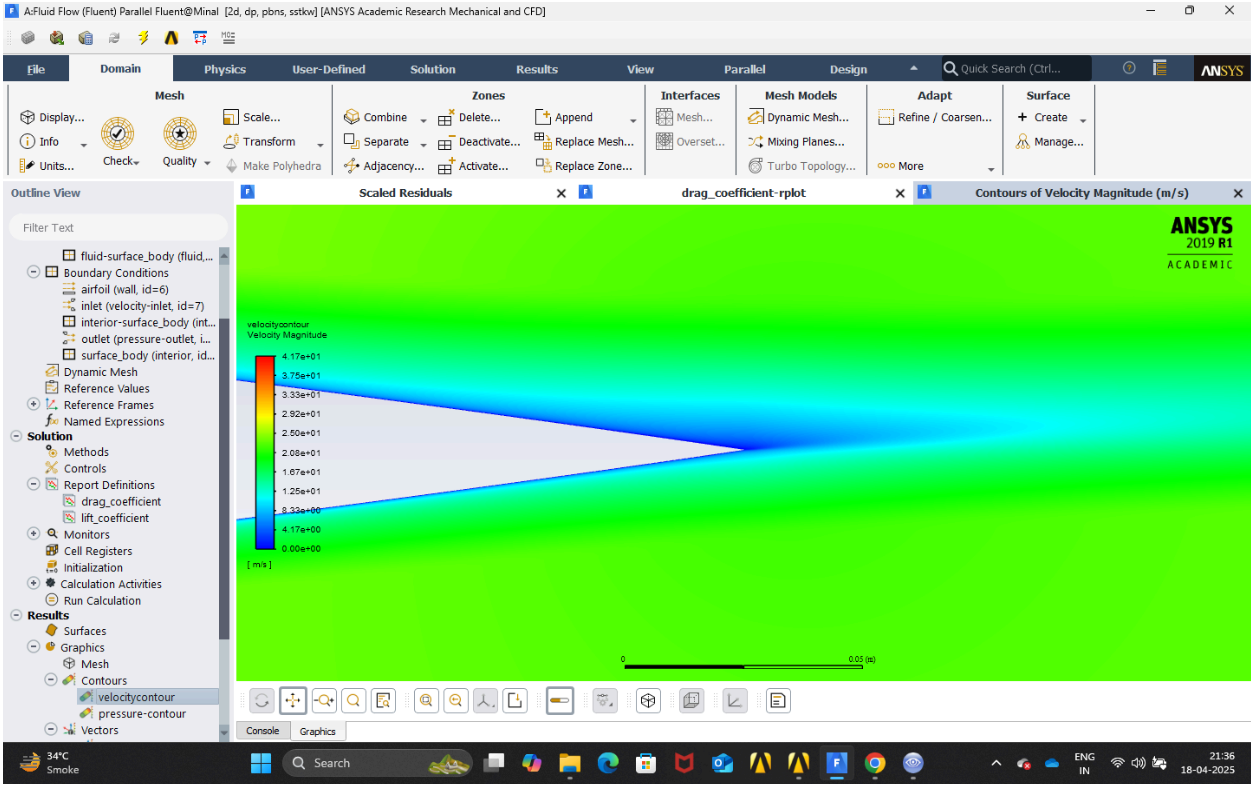
Case - 2 Angle of attack = 5 degrees

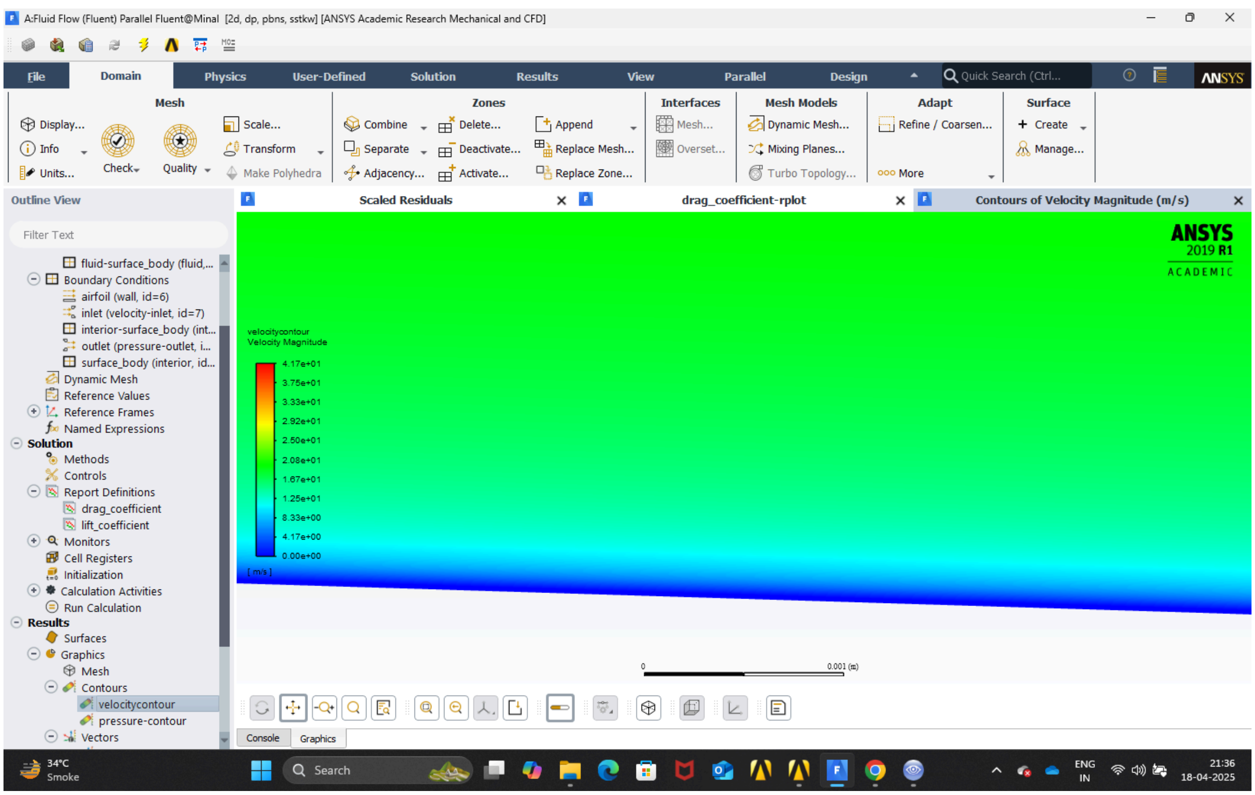
Velocity Contour




A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Contours of Velocity Magnitude (m/s)
velocitycontour
Velocity Magnitude
4.17e+01
3.75e+01
3.33e+01
2.92e+01
2.50e+01
2.08e+01
1.67e+01
1.25e+01
8.33e+00
4.17e+00
0.00e+00
[m/s]
ANSYS
2019 R1
ACADEMIC


A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
Contours of Velocity Magnitude (m/s)
velocitycontour
Velocity Magnitude
4.17e+01
3.75e+01
3.33e+01
2.92e+01
2.50e+01
2.08e+01
1.67e+01
1.25e+01
8.33e+00
4.17e+00
0.00e+00
[m/s]
ANSYS
2019 R1
ACADEMIC


A:Fluid Flow (Fluent) Parallel Fluent@Minal [2d, dp, pbns, sstkw] [ANSYS Academic Research Mechanical and CFD]
File
Domain
Physics
User-Defined
Solution
Results
View
Parallel
Design
Outline View
Scaled Residuals
drag_coefficient-rplot
Contours of Velocity Magnitude (m/s)
ANSYS
2019 R1
ACADEMIC
velocitycontour
Velocity Magnitude
4.17e+01
3.75e+01
3.33e+01
2.92e+01
2.50e+01
2.08e+01
1.67e+01
1.25e+01
8.33e+00
4.17e+00
0.00e+00
[m/s]
Console
Graphics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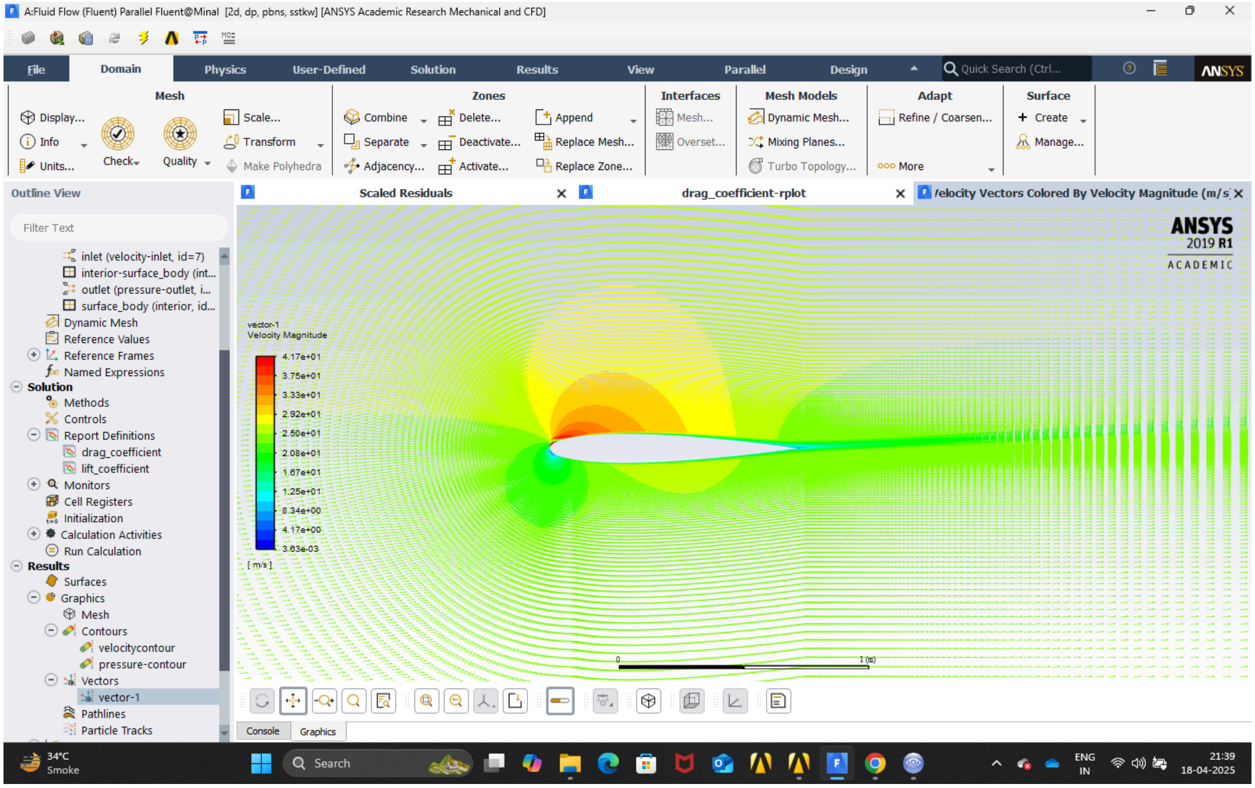
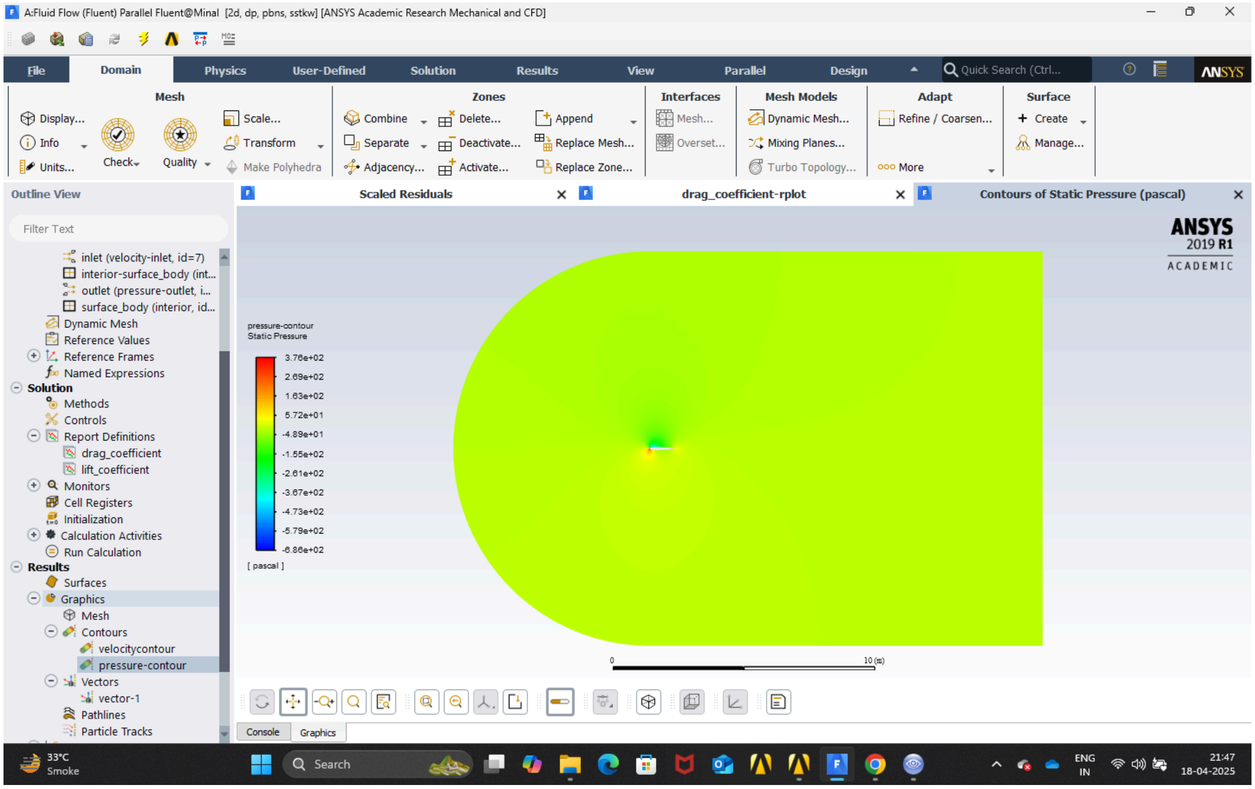
2.08e+01
1.67e+01
1.25e+01
8.33e+00
4.17e+00
0.00e+00
[m/s]
Console
Graphics

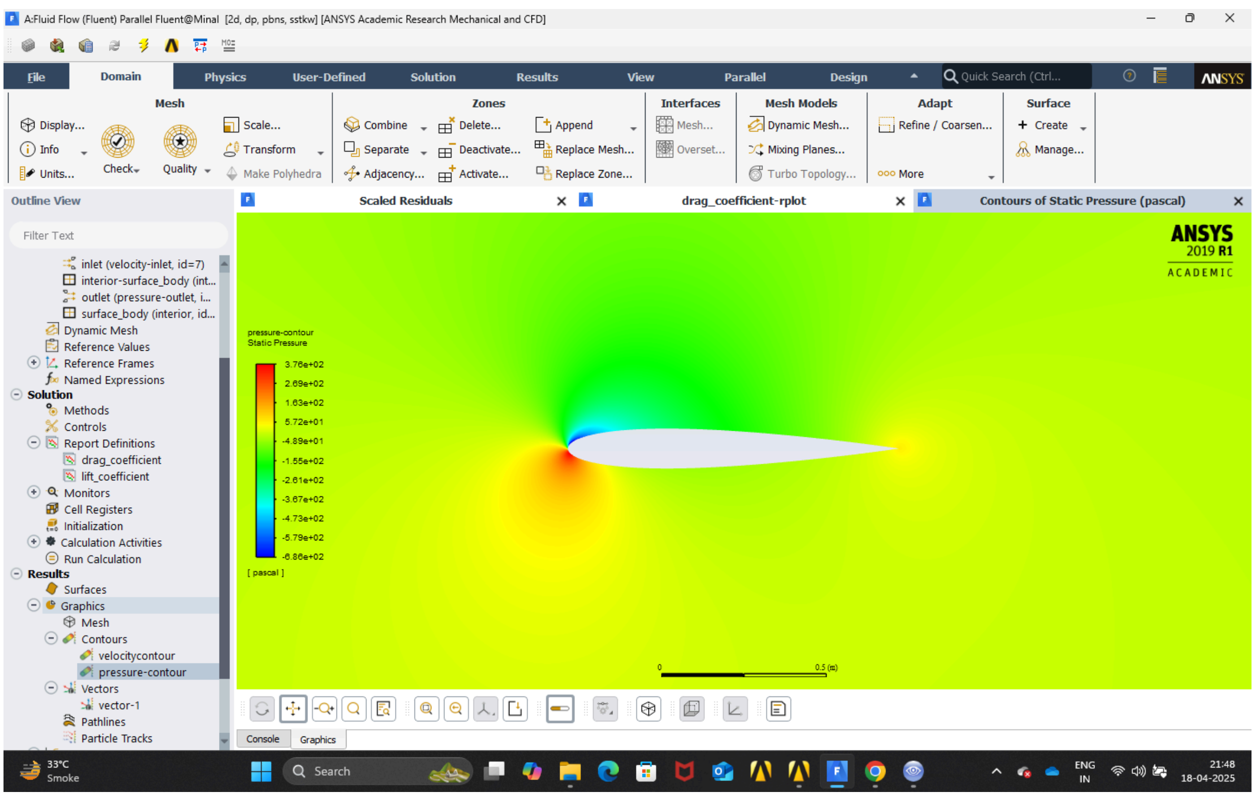


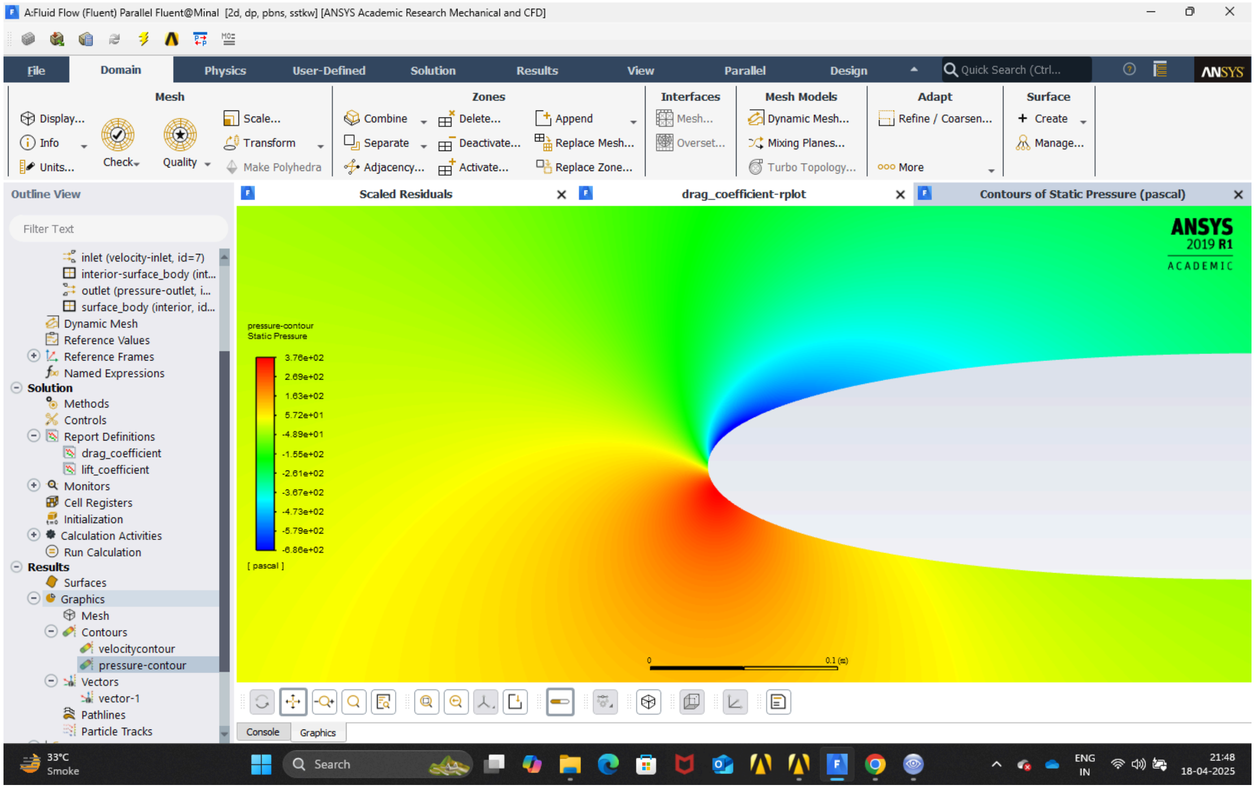
Observations

- The dark blue colour on the walls represents the no slip condition.
- The contours is not symmetric as the angle of attack is 5 degrees,
- The point at the leading edge of the airfoil where the velocity is zero has shifted a little down as the reason is the angle of attack is now 5 degrees so the position of stagnation point has changed.
- In general the velocity above the airfoil is more than the velocity below the airfoil which shows that the lift coefficient will be positive which indeed it is.
- The no slip condition indicated by dark blue colour is still valid.

Pressure Contour



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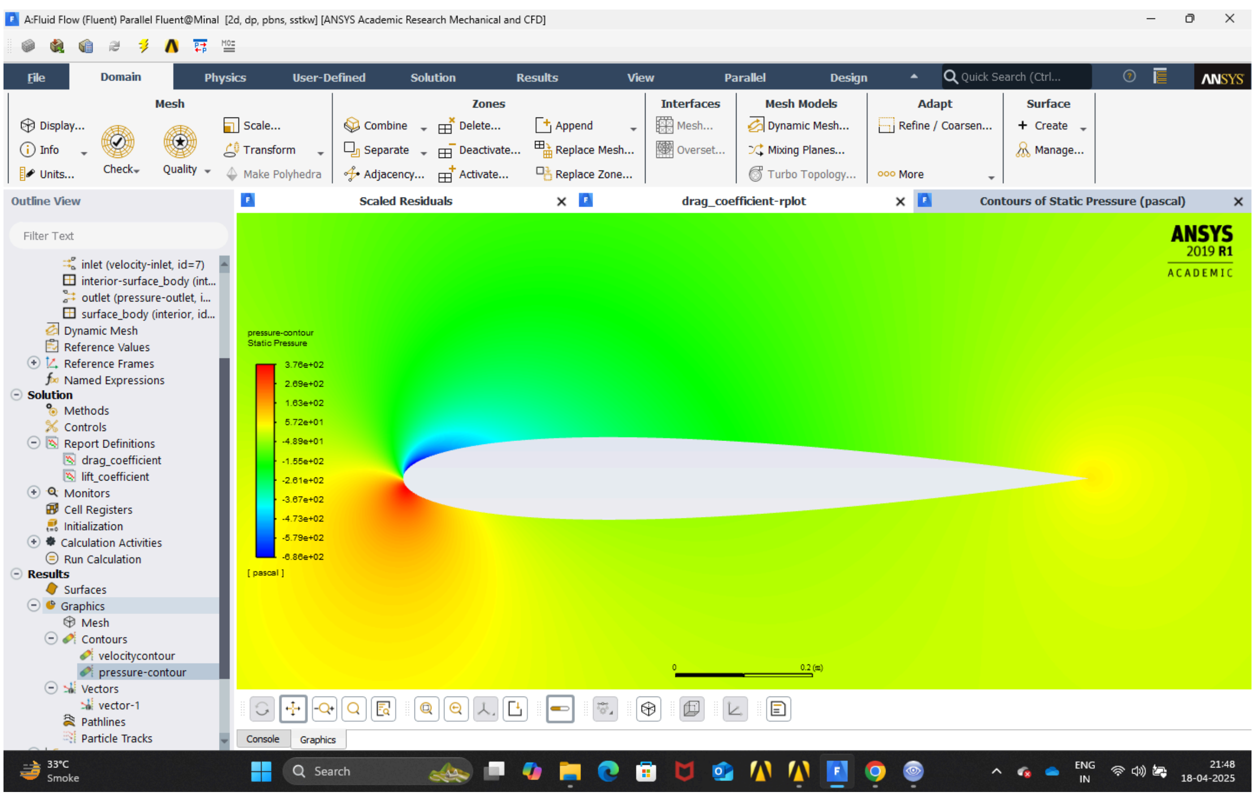
File
Domain
Physics
User-Defined
Solution
Results
View
Parallel
Design
Quick Search (Ctrl...
Mesh
Display...
Info
Units...
Check
Quality
Scale...
Transform
Make Polyhedra
Zones
Combine
Separate
Adjacency...
Delete...
Deactivate...
Activate...
Append
Replace Mesh...
Replace Zone...
Interfaces
Mesh...
Overset...
Mesh Models
Dynamic Mesh...
Mixing Planes...
Turbo Topology...
Adapt
Refine / Coarsen...
More
Surface
Create
Manage...
Outline View
Filter Text
inlet (velocity-inlet, id=7)
interior-surface_body (int...
outlet (pressure-outlet, i...
surface_body (interior, id...
Dynamic Mesh
Reference Values
Reference Frames
Named Expressions
Solution
Methods
Controls
Report Definitions
drag_coefficient
lift_coefficient
Monitors
Cell Registers
Initialization
Calculation Activities
Run Calculation
Results
Surfaces
Graphics
Mesh
Contours
velocitycontour
pressure-contour
Vectors
vector-1
Pathlines
Particle Tracks
Scaled Residuals
drag_coefficient-rplot
Contours of Static Pressure (pascal)
ANSYS
2019 R1
ACADEMIC
pressure-contour
Static Pressure
3.76e+02
2.69e+02
1.63e+02
5.72e+01
-4.89e+01
-1.55e+02
-2.61e+02
-3.67e+02
-4.73e+02
-5.79e+02
-6.86e+02
[pascal]
0
0.1 (m)
Console
Graphics
33°C
Smoke
Search
ENG
IN
21:48
18-04-2025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ANSYS
2019 R1
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pressure-contour
Static Pressure
3.76e+02
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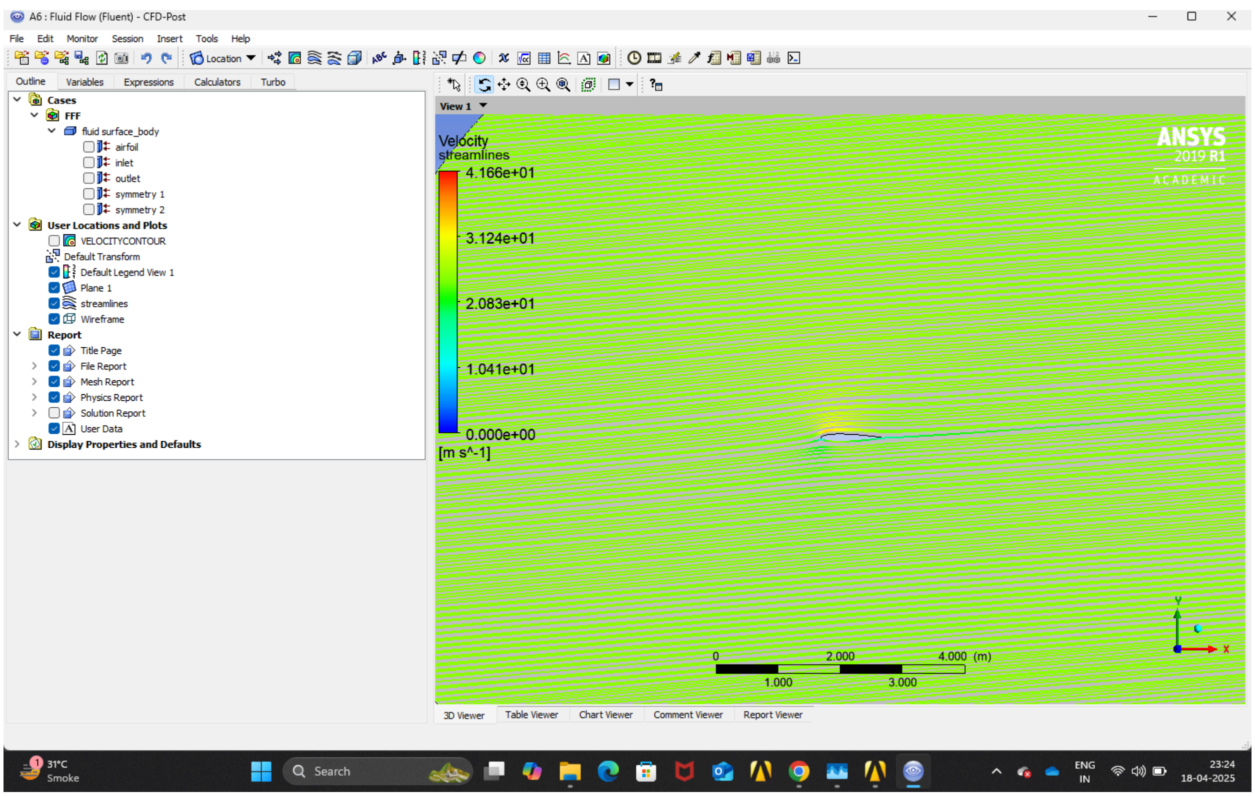
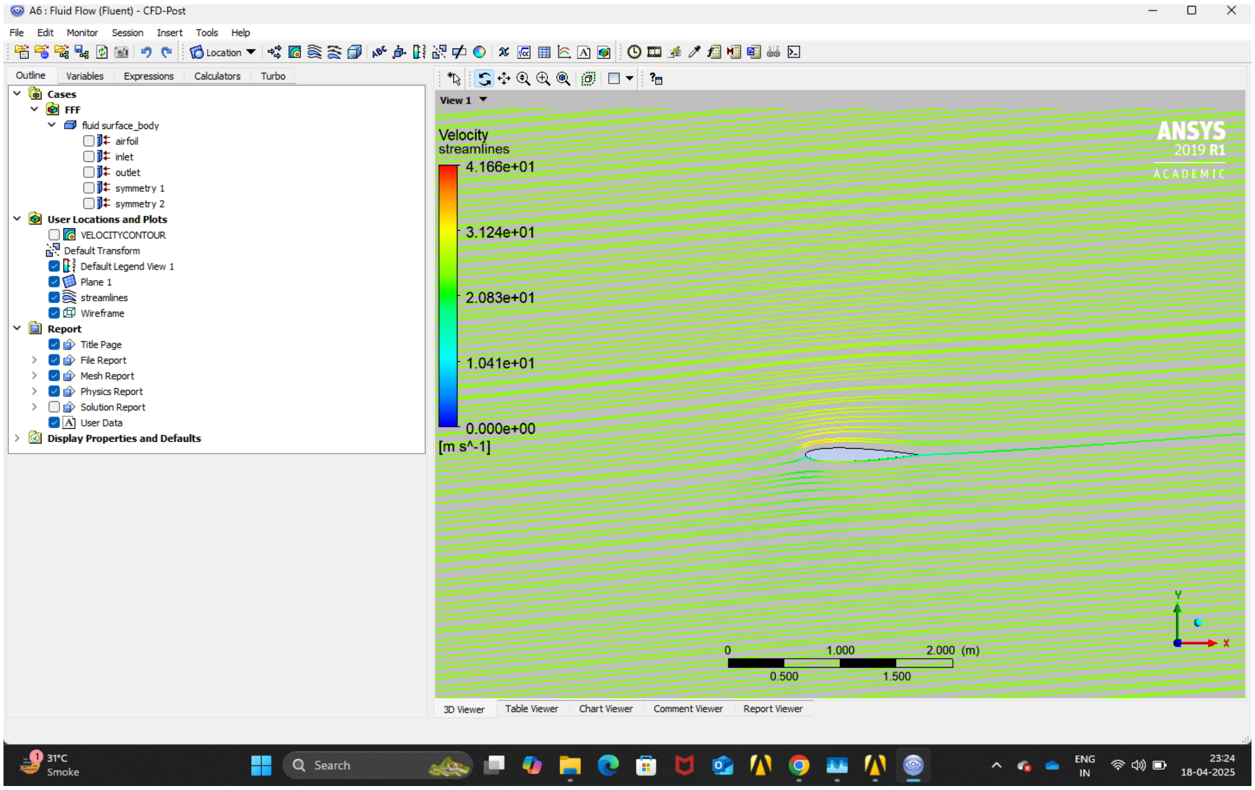
-4.89e+01
-1.55e+02
-2.61e+02
-3.67e+02
-4.73e+02
-5.79e+02
-6.86e+02
[pascal]
0
0.2 (m)
Console
Graphics
33°C
Smoke
Search
ENG
IN
21:48
18-04-2025

Observations

- The pressure is equal to zero at just above the leading edge of the airfoil.
- The pressure is maximum at just below the leading edge of the airfoil.
- The contours are not symmetric as the angle of attack is now 5 degrees.
- The contours are following Bernoulli's principle as in the velocity contour the point where velocity is maximum, the pressure is minimum and vice-versa.
- In general the pressure below the airfoil is greater than pressure above the airfoil signifying positive lift coefficient and lift force.

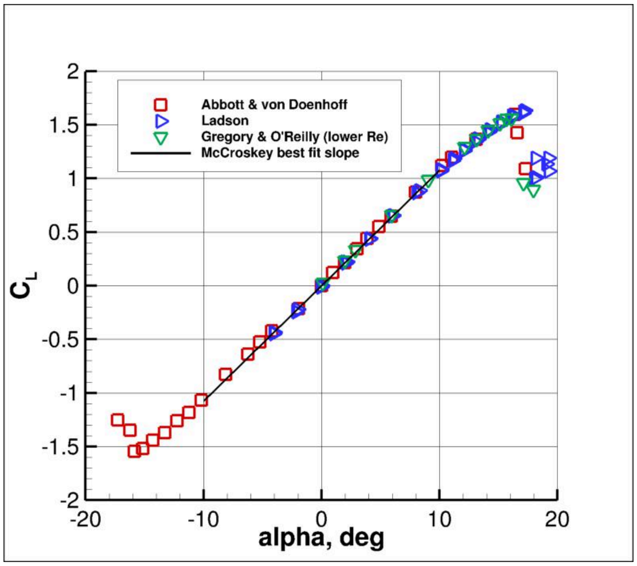
Streamlines



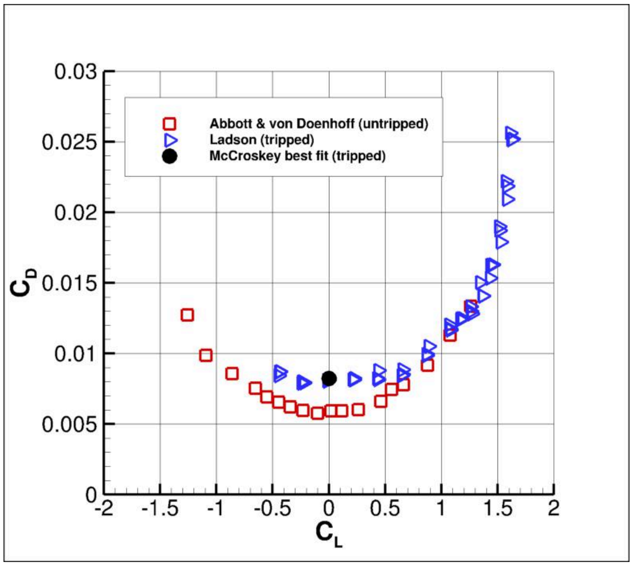


Observations

- The streamlines far away from the airfoil are mostly parallel lines angled at 5 degrees from the horizontal because now the flow is coming at an angle of 5 degrees.
- Near the airfoil, we can observe the streamline flowing over it as the streamlines are curved.
- The velocity increases from the leading edge of the airfoil to a point before the mid of the airfoil and then again decelerates. It is indicated by the streamlines getting to a colour of yellow and then again going back to green.



Graph of AOA with Lift coefficient



Graph of Lift coefficient with Drag coefficient

Angle of Attack	Parameters	Ansys - CFD results	Literature
0^o	C_D	0.010069	0.008
	C_L	6.9584 E-09	0
5^o	C_D	0.011565	0.009
	C_L	0.52867	0.544

Observation of variations of lift and draft coefficients with angle of attack.

- As the angle of attack changes from 0 to 5 degrees the lift coefficient changes from a value of 6.9584 E-09 to 0.52867 signifying that the lift force at zero degree will be zero and the lift force at 5 degree will not be zero which was also observed during the simulation as the lift force that I got for zero degree angle of attack was 2.6554E-6 N which is approximately equal to zero and it changed to 198.54N at an angle of attack of 5 degree, this result is also expected as the results of velocity and pressure contours explain this very well.

- As the angle of attack changes from 0 to 5 degrees the drag coefficient changes from a value of 0.010069 to 0.011565 signifying that the drag force at zero degree will be quite less and the lift force at 5 degree will be greater than 0 degrees this result is also expected because as the angle of attack increases the area facing the flow increases which means more flow will be restricted increasing drag force.

Conclusion

The results from the simulation match quite accurately with the results of the literature, which suggests that the setup used is correct. The lift increased with increase in angle of attack which is also confirmed from the increased lift coefficient and the velocity and pressure contour also explains the results obtained similarly the drag coefficient also increased as the angle of attack which match the literature results it is also self explainable because the frontal area exposed to the flow increases with increase in angle of attack.

Also, there was no flow separation observed for 0 degree or 5 degree angle of attack, meaning that pressure drag in these cases will be negligible, but as the angle of attack increases and goes beyond a threshold value, flow separation will start to occur which will increase the pressure drag and the consequently the drag coefficient will also increase.

Acknowledgement

I would like to sincerely thank Mr. Ravi Teja sir for clarifying my doubts regarding Ansys Fluent and helping me understand the working CFD simulations.

References

- https://youtu.be/3i9Ryq-m1HA?si=5BAvwQ1W2kD3JR36
- https://www.me.psu.edu/cimbala/Learning/ANSYS/Workbench_Tutorial_Airfoil.pdf
- https://images.app.goo.gl/NyQ4TnFqT5qt9PGEA
- https://turbmodels.larc.nasa.gov/naca0012_val.html
- https://help.sim-flow.com/validation/naca-0012-airfoil